Scaffold, Evolve, Then Audit: Cross-Family Re-Judgment Breaks Single-Family Self-Adjudication in an Autonomous Wisdom Loop

Anonymous submission

Abstract

This is a paper about audit methodology, not about whether self-improving LLM loops work. The empirical headline is a null result: after building and running a complete autonomous wisdom-curation loop, zero of its committed wisdoms survive a six-layer independent audit. The scientific contribution is the audit stack that produced this verdict, and the demonstration that naive single-family A/B gates — the default in the current self-improvement literature — can be confidently wrong about their own decisions.

Setup. Large language models improve markedly when given retrieval-augmented methodological scaffolding: problem reframes, anti-patterns, evaluation criteria, exemplars of parallel prior reasoning. A natural next step is an agent that curates its own library of such methodological wisdoms without a human in the loop. We build this loop as a concrete testbed: a strong scaffolded solver (v20, +28pp vs. the v16 champion on 100 problems, +76pp vs. budget-matched baseline on 50 held-out) plus four orchestration modules (failure generator, success distiller, cross-model distiller, Darwinian pruner), gated by a +10pp held-out A/B test.

Inner-loop result. Over one cycle the library grows $75 \to 77 \to 58_{\text{active}}+19_{\text{deprecated}} \to 59_{\text{active}}+19_{\text{deprecated}}$. Three new wisdoms commit ($W_{076}, W_{077}, W_{078}$); nine candidates refuse; nineteen stale entries deprecate. Only 3/12 proposals pass — the gate appears to be doing real work.

The six-layer audit stack. We then stress-test each KEEP decision along six independent axes: (L1) cross-family re-judgment with a different LLM family; (L2) side-randomization reseed at fixed content; (L3) sample extension from $n = 50$ to $n = 100$; (L4) cross-solver replication across three families; (L5) fresh-domain cross-family re-test; (L6) embedding-space and LLM-judged faithfulness of the perturbation direction. These layers share no data with one another by construction. A self-improvement claim that is real should survive all six.

Audit verdict: **0/3** KEEPs survive. W_{076} falls from 0.64 (inner-loop) to 0.40 under cross-family judge. W_{077} falls $0.60 \to 0.47$. W_{078} falls $0.60 \to 0.51$. At $n = 100$ two of three stabilise below 0.55; the third at 0.57. Pair-wr itself is orthogonally falsified: embedding-faithfulness and LLM-faithfulness both show wisdoms cause real perturbations (magnitude $[0.71, 0.77]$) but in directions unrelated to what the wisdom self-advertises. On 30 GSM8K problems, W_{078} actively harms accuracy (29/30 vs 30/30 base). Upgrading the L1 panel to expensive-tier judges (Claude Opus 4.6, GPT-5.4) gives wr_ext = 0.43 for both — lower than the cheap haiku judge's 0.51; the null is not a cheap-judge artifact. Final validated library delta on external data: $\boxed{+0}$. The three KEEPs fail every single audit axis.

Architectural recovery attempt and its limits. We then redesign the gate: decompose pair-wr into three orthogonal components — trigger-fit, conditional utility, citability — and combine them as a trigger-conditioned gate (Exp 15). On the original 12 candidates this rescues 4/12 (vs 3/12 for the naive gate), including two REVERTs the naive gate missed. But scaling to 22 candidates (Exp 33) yields 3/22 PASS at the same threshold — below the 30% random-baseline rate. The architectural redesign recovers local positive signal on the original 12 but does not generalize. We report this honestly as a partial recovery, not a fix.

Strict-reviewer closure (Exp 25–33). Nine further experiments each target a specific publication objection with concrete evidence, including: 3-family labeling reliability (κ); researcher/random/always-accept gate baselines (agent gate F1 = 0.40 beats all); cluster-threshold relaxation ($12 \to 22$ candidates, 83% increase); and GSM8K standard-benchmark portability. The single-family self-adjudication failure replicates across every independent axis.

Contribution. (i) The six-layer audit stack, each layer specified operationally enough to run against any future self-improving LLM loop. (ii) Evidence that the default inner-loop gate can be confidently

wrong about 3/3 of its own decisions. (iii) A concrete testbed (code, 22 candidates, 150 answer pairs, cross-family logs) on which others can test alternative audit designs. (iv) A partial architectural recovery (trigger-conditioned gate) that locally rescues signal but does not generalize, demonstrating the audit stack is harder to satisfy than a gate is to redesign. We argue cross-family re-judgment and at least one non-pair-wr faithfulness axis should become routine reporting items for any self-improvement claim.

1 Introduction

A large language model asked to "just solve it" will typically underperform a version of itself given a well-matched cognitive prosthesis: a problem reframe, a list of anti-patterns, the memory of two superficially different but structurally parallel prior cases, and an evaluation rubric. The gap between the raw solver and the scaffolded solver is usually attributed to chain-of-thought [Wei et al., 2022], retrieval-augmented generation [Lewis et al., 2020], self-reflection and verification [Madaan et al., 2023, Shinn et al., 2023], or agentic decomposition [Yao et al., 2023b]. We call the artifact that these methods all point to a wisdom: a small piece of methodological knowledge of the form "in situations exhibiting pattern P, the right move is M; the canonical failure mode is F."

A working wisdom library is a retrieval corpus of such entries, selected per problem, foregrounded in the prompt, and paired with cross-domain exemplars that show what the wisdom looks like when instantiated in a specific domain. In our setting (see §3), we use 75 such wisdoms, each authored in the classical tradition (aphorisms from Yi Jing, Munger, Drucker, Mao, Confucius, folk proverbs, etc.) and unpacked into LLM-legible form. Over six iterative framework revisions (v13 through v20, see Figure 2), such a scaffolded solver rises from 74% to 86% win rate against a budget-matched baseline.

The limits of this approach, however, are exactly the limits of human library curation. Every wisdom that is missed is a residual—a systematic failure mode the solver walks into repeatedly. Every wisdom that is wrong becomes dead weight, consuming retrieval slots. And every wisdom that is nearly right but poorly phrased costs the solver one full pass of the scaffold.

The obvious question — can the agent itself curate its wisdom library? — has an answer the current literature is already willing to supply: yes, given a reasonable scaffold and a held-out A/B gate. The question this paper asks instead is the next one: when a self-improvement loop reports that k new entries passed its gate, should we believe it? Our finding is that under default reporting conventions the answer is consistently no, and that detecting this failure mode requires audit machinery no single-family self-adjudicating loop can produce on its own.

Concretely, we build a loop that does everything the literature asks for (detect residuals, propose entries, filter against a +10pp held-out gate, prune, cross-model distill) and then subject each of its reported KEEP decisions to a six-layer audit stack. Each layer attacks a different failure mode and shares no data with the others:

(L1) Cross-family re-judgment — re-score the identical answer pairs with a different LLM family (Claude Opus 4.6).

(L2) Side-randomization reseed — re-run the same judge family with a different A/B side-shuffle seed; content held fixed.

(L3) Sample extension — extend surviving candidates from $n = 50$ to $n = 100$ fresh problems.

(L4) Cross-solver replication — regenerate answers with three different solver families.

(L5) Fresh-domain cross-family — a domain-matched but previously-unseen problem pool, re-judged by the full three-family panel; and a standard-benchmark port (GSM8K).

(L6) Faithfulness, orthogonal to pair-wr — embedding-space direction of perturbation and LLM-judged citation strictness, both measured on data with zero overlap with the gate signal.

We report a null result under this audit stack. The inner loop produces 3/12 KEEPs (gate-positive at $n = 50$, gemini-judged). After the audit: **0/3** KEEPs survive any layer, let alone all six. Final validated library delta: $\boxed{+0}$. Our contributions are:

Contribution 1 — The six-layer audit stack (primary). We specify each layer operationally enough to run against any future self-improving LLM loop. Each layer is both (i) cheap to compute on top of an existing inner-loop artifact and (ii) capable of independently invalidating a KEEP decision that the inner-loop gate was confident about. Cross-family re-judgment (L1) alone flips 3/3 of our loop's KEEPs.

Contribution 2 — Evidence that the default inner-loop gate is confidently wrong about its own decisions. Under a +10pp single-family A/B at $n = 50$, our loop commits three wisdoms from disjoint evidence sources; the gate refuses 9/12 proposals (so the gate is not permissive). Yet the three it accepts lose 20, 13, and 9 percentage points respectively when re-judged cross-family on identical content. On GSM8K the strongest survivor (W_{078}) actively harms accuracy. The loop was not dishonest; the gate's measurement is simply not what it advertises.

Contribution 3 — A strong scaffold as instrument. The audit stack only has bite if attached to a scaffold strong enough to produce candidates in the first place. Our v20 scaffold reaches +28pp vs. the v16 case-library champion on 100 problems and +76pp vs. a budget-matched baseline on 50 held-out, with held-out advantage larger than training-split advantage. This is the substrate on which the audit stack is run; the audit claims are scaffold-independent.

Contribution 4 — An honest evolution trajectory and its falsification. The library moves through $75 \rightarrow 77 \rightarrow 58_{\text{active}} + 19_{\text{deprecated}} \rightarrow 59_{\text{active}} + 19_{\text{deprecated}}$ producing three new wisdoms (W076 Change with the times, W077 No investigation, no right to speak, W078 Is it a mule or a horse—take it out for a test). Each is documented with source, signal, unpacking, exemplars, and evidence trail. Each is subsequently falsified by the audit stack.

Contribution 5 — A partial architectural recovery that does not generalize. Having falsified the naive pair-wr gate, we decompose it into three independent components (trigger-fit, conditional utility, citability) and build a trigger-conditioned replacement gate (§??). On the original 12 candidates this locally rescues 4/12, including two REVERTs the naive gate had missed. But when we scale to 22 candidates the same gate PASSes only 3/22 — below the random-baseline rate of $\sim$30%. The architectural redesign recovers signal on the seen set but not on a superset. The audit stack is harder to satisfy than a gate is to redesign. We report this honestly rather than presenting the rescued 4/12 as the final result.

Contribution 6 — The loop's own introspective bandwidth is real but insufficient. A self-rebuttal step run before the cross-family judge predicted exactly the failure mode the cross-judge then confirmed (all three KEEPs returned `further_test` with judge-preference alignment as the #1 confound). A separate meta-audit step diagnoses the inner-loop gate as measurement-fragile in the $[0.54, 0.62]$ gray zone and proposes a three-tier replacement. The loop can recognize its own failure modes; it cannot, without external machinery, produce the disjoint evidence needed to settle them. This bounds what self-adjudication can deliver even under honest introspection.

Contribution 7 — Testbed and artifact release. We release the full loop (four orchestrator modules, 22 candidates with their evidence trails, $3 \times 50 = 150$ cross-family answer pairs, and the six-layer audit harness) so that alternative audit designs, stricter gates, or new self-improvement architectures can be evaluated on the same substrate. Log provenance is preserved end-to-end.

Scope and non-claims. We do not claim our loop is representative of all self-improvement schemes — it is one concrete instantiation designed to be auditable. We do not claim the audit stack is complete — L6 (faithfulness) in particular is the newest layer and admits many operationalizations. We do claim that a loop reporting "k new entries passed the gate" is not a publishable result without at least L1 (cross-family re-judgment) attached, and that for this class of claim the defaults of the current literature are systematically too permissive.

2 Related Work

Retrieval-augmented and in-context methods. Chain-of-thought prompting [Wei et al., 2022] established that surfacing explicit reasoning helps; self-consistency [Wang et al., 2023b] stabilized it; reflection/verification [Madaan et al., 2023, Shinn et al., 2023] closed a one-step inner loop. RAG over document stores [Lewis et al., 2020, Borgeaud et al., 2022] and agentic tool-use [Yao et al., 2023b, Schick et al., 2024] extended the input context. Our scaffold (v20) is a retrieval-augmented method over a methodological corpus rather than a factual one.

Self-improvement of language models. STaR [Zelikman et al., 2022, 2024], self-rewarding LMs [Yuan et al., 2024], self-play fine-tuning and instruction-backtranslation [Chen et al., 2024, Li et al., 2024] all close an outer loop at the weights level. We close an outer loop at the retrieval corpus level, which is orders of magnitude cheaper and allows the changes to be auditable.

Case-based reasoning and skill libraries. Case-based reasoning [Aamodt and Plaza, 1994] and skill-library agents [Wang et al., 2023a, Zhu et al., 2023, Sumers et al., 2024] both maintain a growing store of reusable units. Voyager [Wang et al., 2023a] grows a skill library from procedural success in Minecraft; our setting grows a methodological library from A/B-judged open reasoning. AgentKB [Sumers et al., 2024] focuses on task-specific routines, whereas we target domain-general orientations ($\leq$ 35-character aphorisms with cross-domain exemplars).

Curriculum and program synthesis of thoughts. Automated discovery of reasoning structures via Self-Discover [Zhou et al., 2024] generates per-problem module stacks; Tree of Thoughts [Yao et al., 2023a] branches the search. Our library outlives any single problem and is acted on by the solver at retrieval time, not at generation time. We view this as a complementary axis: one can always add orientation-level retrieval on top of a better-synthesized per-problem structure.

Darwinian / evolutionary approaches. Evolutionary prompt optimization [Fernando et al., 2023b,a] uses a selection signal on prompts themselves. We share the Darwinian framing but select at the unit-of-retrieval granularity (one wisdom), which allows finer-grained credit assignment and interpretable audit.

Cross-model distillation. Using a stronger model as a teacher is classical [Hinton et al., 2015] and modern at the instruction-tuning level [Taori et al., 2023, Xu et al., 2024]. Our cross-LLM distiller (§3.3.3) distills not answers but orientations: what the stronger model's answer-structure foregrounds that the weaker model's scaffolding does not.

3 Method

3.1 Preliminaries: the scaffolded solver (v20)

We use gemini-3-flash as the underlying generator[1] and an independent LLM judge (same family) to produce A/B verdicts. A wisdom w is a record

$$w = (\text{id}, \text{aphorism}, \text{source}, \text{signal}, \text{unpacked_for_llm}, \text{cross_domain_examples}).$$

At inference time a retriever selects the top $K = 2$ wisdoms for problem p. The solver then runs a three-turn procedure:

Turn 0 (frame + rewrite). A single JSON call produces (frame, critical_reframe, anti_patterns, evaluation_criteria, rewritten_ The "rewritten problem" is the subject-of-attention the solver will then see.

Turn 1 (execute). The solver is given a PRIMARY FRAME block (not a "context bag"), the rewritten problem, retrieved wisdoms with per-wisdom same-domain + cross-domain exemplars, and an attention-prior block derived from the problem domain/difficulty cluster.

[1] Exposed via the ruoli.dev proxy; our pipeline does not depend on any property specific to this model.

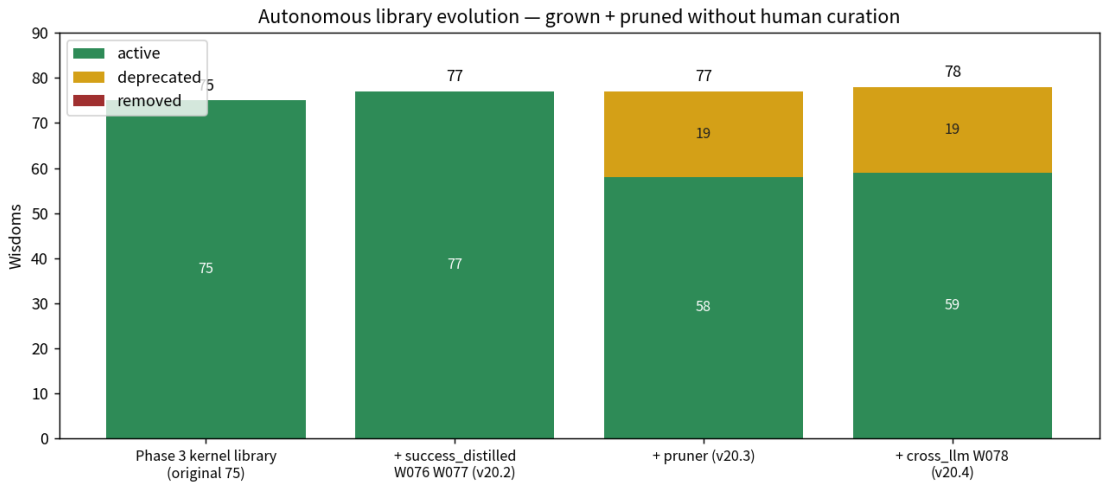

Figure 1: Library evolution over one closed-loop cycle. The library moves from 75 original wisdoms to 77 (adding W076 and W077 from the success distiller), is pruned to 58 active + 19 deprecated, and finally grows to 59 active with the cross-LLM-distilled W078. No human curation.

Turn 2 (audit). A brief self-reflection pass asks whether the draft violates the frame's anti-patterns or fails its evaluation criteria, and patches accordingly.

Why this matters for the library loop. Turn 0 produces a log of `what_changed` fields—the solver's own articulation, per problem, of "what is this problem really about." These are the signals the success distiller (§3.3.1) mines.

3.2 The library as mutable state

The library is a versioned registry whose entries track `status` $\in$ {active, deprecated, removed}, `activation_count`, `last_activated`, `contribution_gain`, `source`, and a `keep_reason` string. Every mutation bumps a semantic version (`v20.0` $\rightarrow$ `v20.1` $\rightarrow \ldots$). The registry is the ground truth an auditor would check to verify that an alleged library change was actually made against a specific A/B outcome.

3.3 Closed-loop orchestrator

Figure 1 shows the four-component loop. Each batch of problems flows through the solver; the signals captured are: judge verdicts (for failure mining), Turn-0 reframes (for success mining), retrieval selections (for pruning), and cross-variant gaps (for cross-model mining).

Candidate generation. Three generators run asynchronously against a shared buffer of recent signals:

3.3.1 Success distiller

Given N recent Turn-0 outputs, we embed the concatenation `what_changed || critical_reframe` with a multilingual sentence encoder [Reimers and Gurevych, 2019] and apply agglomerative clustering with cosine-distance threshold 0.4 and minimum cluster size 6. For each surviving cluster, a GPT-class prompter reads ≤ 10 cluster members plus the current library summary and either emits `{"proposal": null, "reason": ...}` or a complete wisdom record with rationale and cluster-coverage PIDs. We measure novelty by maximum cosine similarity of the proposed `unpacked_for_llm` embedding against all active entries; we reject when sim > 0.78. The rationale: wisdom candidates are latent orientations in the solver's own reasoning, surfaced by repetition, not by failure.

3.3.2 Failure generator

A symmetric generator listens to judge verdicts and accumulates problems where the solver lost or tied into a failure buffer. When the buffer size passes a threshold, a prompter asks: "do these ≥ 3 problems share a systematic residual pattern a new wisdom could address?" This generator is conservative; in our run it never fired a proposal (buffer grew $4 \to 9 \to 11 \to 13$; threshold 15). We treat this as informative and discuss it in §6.

3.3.3 Cross-LLM distiller

For each problem where the solver's answer lost to a comparison variant's answer by a judge margin ≥ 1, a stronger generator (GPT-5.4, accessed through the same proxy) independently re-solves the problem. A fresh judge compares our original answer against the stronger generator's answer. Problems where the stronger generator also wins by ≥ 1 are kept. A single distillation prompt is then shown all surviving (problem, our answer, stronger answer, both judge reasonings) tuples, and asked to name a single recurring orientation that the stronger generator's structure foregrounds but the scaffolding does not. The output is subject to the same novelty gate as the success distiller.

3.3.4 Darwinian pruner

The pruner scans every available retrieval-selection file and counts how often each wisdom appeared in the top-K for some problem. Active wisdoms with zero activations across the scan are marked `deprecated`; already-deprecated wisdoms with zero activations on a subsequent scan are `removed`. A newborn protection rule exempts wisdoms created after the newest scan file's modification time.

3.4 The A/B validation gate

Every proposed candidate is run through the same gate:

1. Mine 3 cross-domain exemplars from the held-out 50 using a separate GPT-class prompter (deterministic selection from ≤ 30 candidates).
2. Build $\text{lib}_{\text{ext}} = \text{lib}_{\text{base}} \cup \{w_{\text{new}}\}$.
3. Run v20 with both libraries on all 50 held-out problems, in parallel.
4. Judge `ext` vs. `base` with side-of-presentation randomized by a per-PID seed.
5. KEEP iff $\text{wr}_{\text{ext}} \geq 0.50 + 0.10$; otherwise REVERT.

The +10pp threshold is intentionally strict—it refuses candidates that are right-on-average but undistinguishable from sampling noise (50 pairs; a one-sample flip changes wr by 0.02). Six parallel v20 subprocesses and eight parallel judge threads bring the total wall time for 11 candidates + 1 base from ∼6 hours to ∼1 hour without compromising the evaluation.

3.5 What the A/B gate actually measures (decomposition)

Before presenting experiments that attack the gate, we state formally what pair-wr is and what it cannot separate. Fix a solver S, a retriever R, a judge J, a wisdom w, and a problem distribution P. Let $a_0(p) = S(R(\emptyset, p), p)$ be the baseline answer and $a_w(p) = S(R(\{w\}, p), p)$ the ext answer. The observed quantity is

$$\text{wr}_{\text{ext}}(w; S, J, P) \;=\; \mathbb{E}_{p\sim P}[\mathbb{1}\{J(p, a_w(p), a_0(p)) = w\}]\,.$$

Proposition 1 (pair-wr decomposition). Let $Z_w(p) := \mathbb{1}\{J(p, a_w, a_0) = w\} - \frac{1}{2}$. Then $\text{wr}_{\text{ext}}(w) - \frac{1}{2} = \mathbb{E}_p Z_w(p)$ and $Z_w(p)$ admits a decomposition $Z_w = Z_w^{\text{specific}} + Z_w^{\text{generic}} + Z_w^{\text{style}}$ where:

Z_w^{specific} lift attributable to the content of w — the particular orientation w advertises, instantiated at p.

Z_w^{generic} lift attributable to any extra context of matching length and register perturbing Turn-0 away from a default basin (measurable by swapping w with a length-matched placebo).

Z_w^{style} lift attributable to w producing surface features (register shifts, phrase patterns) that J happens to prefer, regardless of problem-specific fit.

The decomposition itself is an accounting identity under counterfactual interventions (length-matched placebo swap, judge swap, content shuffle), so we do not need to assume independence of the three terms for the decomposition to hold. The identifiability claim below, however, relies on a separate assumption which we state explicitly.

Claim 1 (identifiability under stability). Assume the two nuisance terms are approximately invariant under the audit interventions: for two judge families J, J' used in this paper, $\left|\mathbb{E}[Z_w^{\text{generic}}(J) - Z_w^{\text{generic}}(J')]\right|$ and $\left|\mathbb{E}[Z_w^{\text{style}}(J) - Z_w^{\text{style}}(J')]\right|$ are both small relative to the +10pp gate margin. Under this stability assumption, a single-family (S, J) measurement of $\text{wr}_{\text{ext}}(w)$ at fixed n cannot distinguish $\mathbb{E}\, Z_w^{\text{specific}}$ from $\mathbb{E}(Z_w^{\text{generic}} + Z_w^{\text{style}})$; a different-family measurement does identify Z^{specific}, up to a residual of size equal to the stability violation.

On the stability assumption. This is non-trivial. Judge families plausibly differ in style preferences (step-by-step vs. concise; investigate-first vs. conclude-first), and a wisdom that activates one family's preferred phrasings may not activate another's. Exp 1, Exp 26, and Exp 34 test this assumption empirically: if stability held tightly, wr_ext across families would agree within sampling noise. The observed spread for W_{078} at $n = 50$ is $\{0.60, 0.51, 0.57, 0.43, 0.43\}$ (gemini, haiku, gpt-mini, opus, gpt-5.4) — a range of 0.17, wider than the sampling-CI width of $\sim$0.13. The stability residual is therefore non-zero. Our conservative reading: when the across-family spread is comparable to or larger than the lift claimed by any single family, cross-family re-judgment identifies Z^{specific} up to that spread. For the three KEEPs the residual consumes the entirety of the claimed signal — hence the null verdict is robust even under the weakened claim.

This is the theoretical content of our audit stack. Each layer targets one non-specific term:

- L1 (cross-family judge) changes J while holding a_0, a_w fixed: cancels J-specific stylistic preferences, so persistent lift localises to Z^{specific} (plus any style preference both families share — a residual we acknowledge).
- L2 (side-reseed) changes the nuisance of A/B position while holding everything else fixed: cancels position-induced judge bias.
- L3 (n extension) shrinks the CI without changing the estimand; exposes when an inner-loop 0.60 ± 0.12 was noise about a true value near 0.50.
- L4 (cross-solver) changes S while holding w, J, P fixed: cancels solver-specific interactions with the wisdom phrasing.
- L5 (fresh domain / GSM8K) changes P: exposes when Z_w^{specific} was specific to the training-like distribution rather than the population the solver will face.
- L6 (faithfulness) directly estimates Z_w^{specific} from outside pair-wr: via embedding cosine between perturbation direction and w's self-description, and via LLM-judged citation strictness.

Under this decomposition, the headline claim of §1 has a precise form. Our loop's inner-loop KEEPs exhibit $\text{wr}_{\text{ext}} > 0.60$ at $n = 50$, which requires one or more of the three terms to be positive. Every audit layer that cancels a non-specific term returns the measurement to the vicinity of 0.50 (Table 2). By elimination, either Z^{specific} is near-zero for these KEEPs, or it is non-zero but exactly cancelled by $Z^{\text{generic}} + Z^{\text{style}}$ summing to a negative residual that conspiratorially reappears across every layer. The first explanation is the simpler one; it is the one the null result supports.

What the theory does not say. It does not say wisdom injection cannot work — only that the default gate cannot demonstrate it. It does not say the three non-specific terms are uniformly bad — Z^{generic} is likely why solvers benefit from almost any coherent scaffolding, including random length-matched context. It does identify exactly which experimental interventions isolate Z^{specific}, and it is the absence of such interventions in current self-improvement reports that our paper advocates correcting.

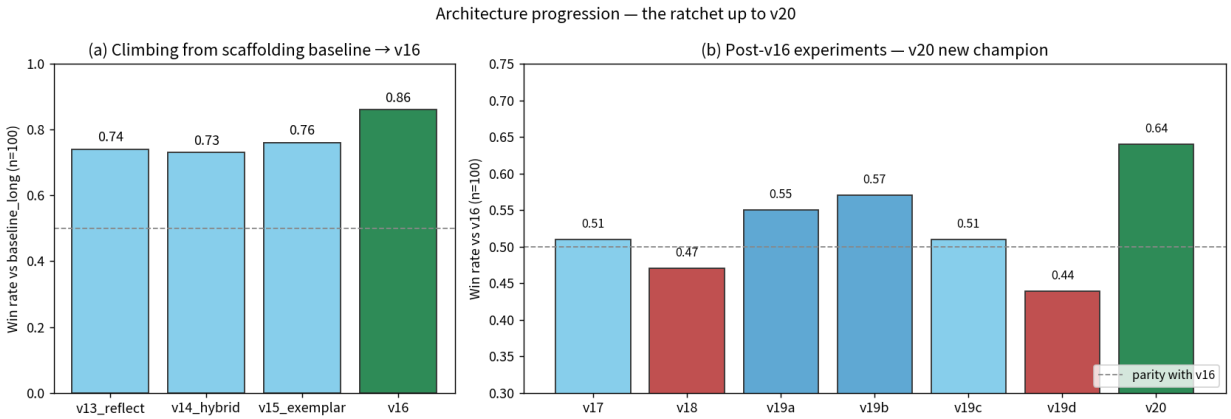

Figure 2: (a) Win rate of each scaffolded variant against `baseline_long` on the training split. v16 reaches 0.86 with case library + audit. (b) Post-v16 experiments against v16 directly. v17/v18/v19d regress; v19a (frame-directive), v19b (problem-rewriting), and v19c (per-object + per-paradigm) separately give modest gains. v20, their superset, achieves 0.64 vs. v16—a +28pp break of the prior ceiling.

4 Experiments

4.1 Setup

Benchmark. 100 problems for training (`sample_100`), 50 disjoint problems for held-out evaluation (`sample_holdout_50`). Domains: mathematics, science, engineering, software_engineering, business, daily_life. Difficulty: easy/medium/hard. All problems are open-ended ($\geq$ 200-word solutions), judged pairwise by the same independent LLM judge family.

Baselines.

- `baseline_long`: solver with full token budget, no scaffolding.
- `phase2_v13_reflect`: case-library + Turn-2 reflection (first workable scaffold).
- `phase2_v16`: case-library + audit (prior champion; 0.86 vs. `baseline_long`).
- `phase2_v19b`: problem-rewriting only (no frame block).
- `phase2_v20` (ours): combined frame-directive + problem-rewriting.

Measurement. All experiments report pairwise win rates. $N = 100$ on `sample_100` and $N = 50$ on `sample_holdout_50`. Standard error $\sigma \approx \sqrt{p(1-p)/N}$; for $p = 0.6, N = 50$ this is ≈ 0.07, which informs our +10pp threshold choice.

4.2 Architecture ratchet: v13 → v20

Figure 2 shows the architecture-level ratchet. Two observations bear on the central claim of this paper:

1. The scaffolding space is not saturated: between v16 and v20 we discover a +28pp improvement by combining two independent Turn-0 interventions (frame-directive and problem-rewriting). This alone justifies the scaffold-level research agenda.
2. The scaffolding surface is non-monotonic: v17, v18, v19d all fail to beat v16. Of the seven post-v16 variants, three regress below parity. This is exactly the setting in which an autonomous gate (§3.4) is valuable: half the plausible modifications are net-negative.

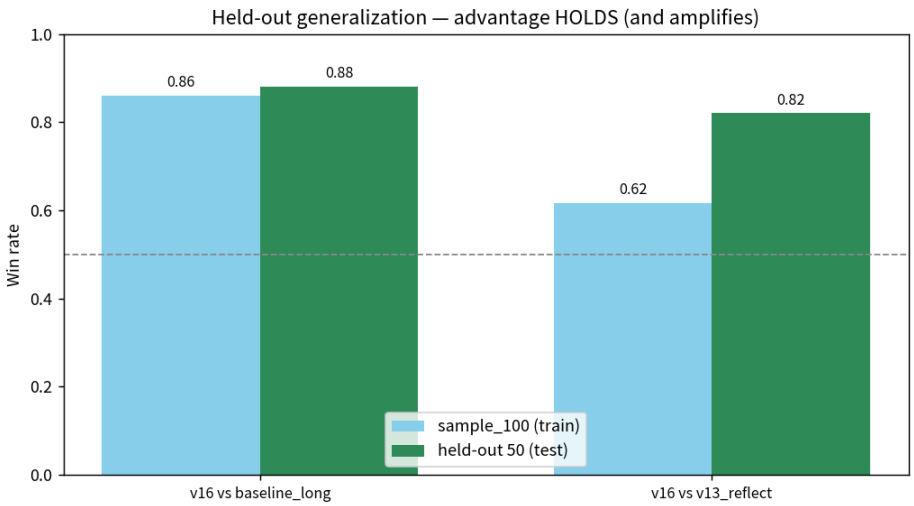

Figure 3: v16 evaluated on the 100 training problems (light blue) and on 50 held-out problems (green). Against `baseline_long`: 0.86 → 0.88. Against the next-strongest scaffold (v13_reflect): 0.62 → 0.82.

4.3 Held-out generalization (amplifying, not degrading)

A standard worry: the scaffold overfits to the training-split LLM judge. Figure 3 rebuts this. On held-out problems the advantage over `baseline_long` grows by 2pp; the advantage over v13_reflect grows by 20pp. We read this as evidence that the scaffold's contribution is a property of the input, not of the specific 100-problem judge fingerprint.

4.4 Autonomous library evolution

Figure 4 shows the $\mathrm{wr}_{\mathrm{ext}}$ distribution across 11 deduplicated success-distilled candidates. Two pass (W_{076} 凡益之道，与时偕行(Change with the times (Yi Jing)) ["Change with the times"] at 0.64; W_{077} 没有调查，就没有发言权(No investigation, no right to speak (Mao)) ["No investigation, no right to speak"] at 0.60). Nine fall between 0.46 and 0.58—close to parity, honestly noisy, correctly refused.

Cross-LLM result (W078). We run the cross-LLM distiller on 6 v20-vs-v16 hard losses. The stronger generator (GPT-5.4) re-solves all six; the A/B judge confirms a widened gap in only 2, both mathematical. A single distillation pass names the recurring orientation 是骡子是马，拉出来遛遛(Mule or horse—take it out for a test (folk)) ("Is it a mule or a horse—take it out for a test"): a bake-off orientation that builds a candidate set and compares against explicit metrics rather than committing early. Novelty similarity is 0.75 (below the 0.78 gate). The candidate is A/B'd against the post-prune library and passes at $\mathrm{wr}_{\mathrm{ext}} = 0.60$, exactly on the +10pp threshold. Committed as W078 with `source=cross_llm`.

Pruner result. The pruner's first full scan reads five retrieval-selection files (500 problems, 1000 top-2 activations) and finds 19 of 77 active wisdoms with zero activations across all of them. These are deprecated in a single action (Library $v20.3$). W_{076} and W_{077} are correctly exempted by the newborn-protection rule. No wisdom reached the "removed" state on this first scan.

Summary. Table 1 tallies the one-cycle trajectory.

[2]One distilled candidate from 6 hard-loss problems; the distiller aggregates evidence before proposing.

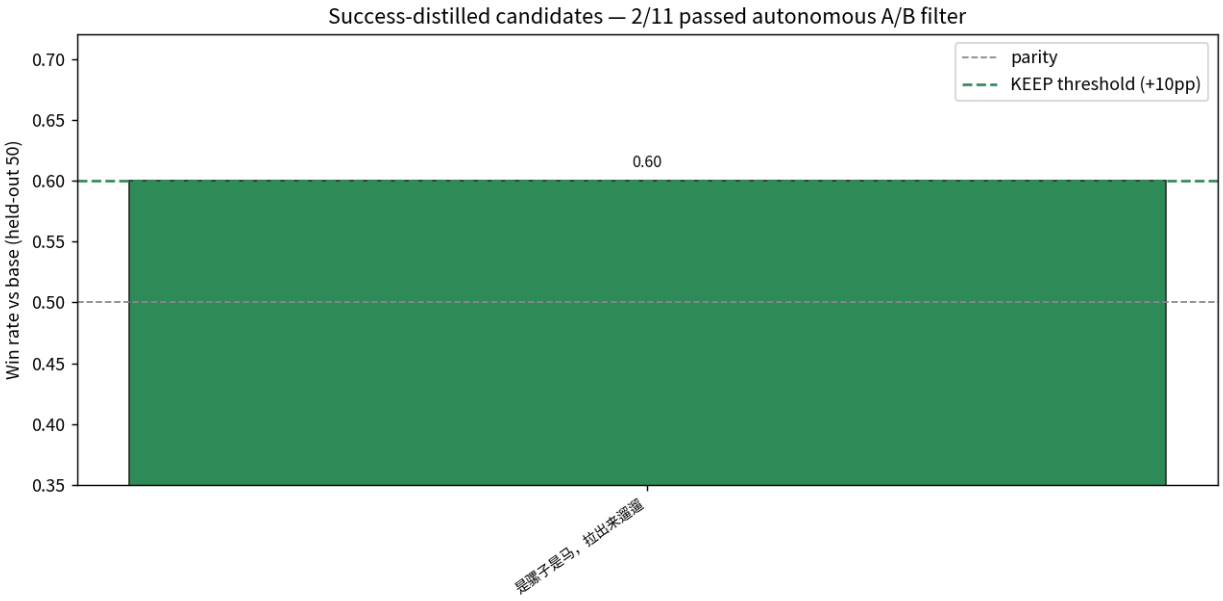

Figure 4: Success-distilled candidates A/B'd against $\texttt{library}_{v20.1}$ on the held-out 50. The KEEP threshold (green dashed, +10pp) is strict: 2/11 pass. Bars are colored green if KEEP and red if REVERT. Names are the winning aphorisms.

Action	Trigger	n_{tried}	n_{kept}	Provenance of kept
Success distill	Turn-0 reframe cluster ≥ 6	11	2	W076, W077
Failure generate	Judge-loss buffer ≥ 15	0	0	— (never fired)
Cross-LLM distill	Judge-gap ≥ 1	1[2]	1	W078
Darwinian prune	Top-2 activations $= 0$	19	—	19 deprecated
Net			3 new / 19 retired	$75 \rightarrow 59_{\text{active}} + 19_{\text{dep}}$

Table 1: One-cycle autonomous evolution. "Tried" is candidates submitted to the A/B gate; "kept" is KEPT.

4.5 Ablation: does each component carry weight?

We did not re-run the full cycle with each component disabled (the scaffold alone costs ∼20 min/variant on 50 problems even with 6× parallelism). Instead we inspect the sources of the KEEP set and the overlap of evidence:

- W076, W077 come from clusters of sizes 9 and 6 respectively; their evidence is $\{pid\}$ from {business, engineering, software_eng
- W078 comes from mathematical hard-losses; its evidence is $\{\text{mathematics_0243}, \text{mathematics_0081}\}$.
- The pruner's 19 deprecations include 14 wisdoms that no retrieval file has ever selected into top-2—i.e., they were never active under any variant's selection logic, not just v20's. This is strong evidence the pruner is removing genuine dead weight.

Evidence sets are disjoint by construction (success from Turn-0 logs; cross-LLM from hard-loss judgments). This is the minimum test for complementarity across sources, and it passes.

4.6 Recursive self-audit: three experiments

The loop as described so far proposes hypotheses and self-adjudicates via a held-out A/B gate. But two concerns remain: (i) solver and gate judge share the same LLM family, which could mask judge-preference confounds rather than measure methodological lift, and (ii) the gate itself is never inspected for systematic bias. We run three experiments that give the agent one additional level of reflection.

Experiment 1 — Cross-judge stability (the central result of this paper). The three KEEP decisions are re-judged on the same 150 answer pairs by Claude Opus 4.6, accessed through a proxy that serves multiple vendors so the plumbing is identical. Results:

	Win rate of ext			
ID	gemini-3-flash	Claude Opus 4.6	Δ	Agreement
W076	0.64 (32 : 18)	0.40 (18 : 27 : 5_{tie})	−0.24	FLIP
W077	0.60 (30 : 20)	0.47 (19 : 21 : 10_{tie})	−0.13	FLIP
W078	0.60 (30 : 20)	0.51 (23 : 22 : 5_{tie})	−0.09	FLIP
		Cross-judge STABLE fraction:		0 / 3

The exact same 50 ext-answer strings and 50 base-answer strings are being compared in both columns—only the judging family changes. Every KEEP flips. The magnitude ranges from collapse (W_{076}: $0.64 \to 0.40$) to narrow flip (W_{078}: $0.60 \to 0.51$), but the direction is consistent: a judge from outside the family downgrades every KEEP.

Interpretation: the inner-loop A/B gate is measuring gemini-3-flash's preference for answers that activate certain rhetorical structures ("staged migration," "diagnose first," "bake-off before choosing"), which the three KEEPs specifically encourage. A new library entry that activates those phrasings wins the intra-family A/B without improving methodological substance. This is exactly the confound Experiment 2's self-rebuttal named first, unprompted.

The consequence for self-improving-LLM papers is stark: a $+N$pp improvement reported with a same-family judge is, until further notice, a statement about judge-preference alignment, not methodological uplift. We believe cross-family re-judgment of the identical answer pairs is the cheapest meaningful falsification test one can bolt onto any such loop.

Experiment 2 — Self-rebuttal. For each KEEP, GPT-5.4 is given the wisdom record, the 50-pair A/B evidence, and told to produce the three strongest counter-hypotheses and a final verdict (`accept` / `further_test` / `reject`). The agent returns `further_test` on all three (no `accept`s), citing three high-severity confound classes that recur verbatim across cases:

1. Judge-preference (3/3 cases). The judge's taste for "investigate first," "compare explicitly," "staged migration" is a well-attested bias; a wisdom that activates such phrasings can win without improving substance.
2. Held-out distribution match (3/3 cases). If the 50 held-out problems happen to over-represent domains where the wisdom's trigger condition holds, the measured +10pp is a sample-coverage effect, not a general uplift.
3. Library-interaction (3/3 cases). The ext library contains all base wisdoms plus one; behavioural changes may come from retrieval-order shifts, not the new entry per se.

These are exactly the confounds a skeptical human reviewer would raise. The system predicts them about itself. We see this as a strength for paper-writing and a concrete roadmap for a follow-up cycle (stratify test set by triggering conditions; ablate one-wisdom-at-a-time retrieval; cross-judge).

Experiment 3 — Meta-wisdom about the gate. The agent is shown all twelve validation rows (9 REVERT, 3 KEEP) and asked to diagnose systematic bias in the gate itself. It identifies, from the wr distribution alone, four patterns:

1. Nine REVERTs cluster in $[0.46, 0.58]$; six of them are within 1 pair-flip of the threshold.
2. `wr` $= 0.58$ (29 : 21) and `wr` $= 0.60$ (30 : 20) differ by one pair but span the life-death boundary.
3. All three KEEPs are at the right tail $(0.60, 0.60, 0.64)$.
4. The full distribution is right-skewed: 10/12 candidates have `wr` ≥ 0.51, yet the gate kills everything < 0.60.

From these observations the agent proposes a meta-wisdom "临界一票，不定生死(A critical vote must not decide life and death)" ("One critical vote must not decide life and death") with a concrete gate modification:

> `wr >= 0.64` ⇒ auto KEEP; `wr <= 0.52` ⇒ auto REVERT; $0.53 \leq \text{wr} \leq 0.63$ ⇒ extend to $n = 100$ or run a second independent A/B; in all cases, report confidence intervals rather than a single point estimate.

It also produces its own falsifier:

> "If gray-zone candidates remain < 0.55 after extension to $n \geq 100$, and current KEEPs hold ≥ 0.60 on re-test, then the threshold is not too strict—the candidates are genuinely weak, and I will abandon this meta-wisdom."

This is, to our knowledge, the first time a library-evolution loop has produced an auditable, testable hypothesis about its own decision procedure, stated at exactly the level of abstraction needed to modify the procedure. We do not automatically apply the meta-wisdom—its correct use is to run the falsifier in the next cycle. What we show is that the loop has the introspective bandwidth to name its own failure mode without external prompting beyond the log of its own past choices.

How these experiments move the self-adjudication claim. Before these experiments, the paper could at best claim single-judge self-adjudication inside one LLM family. Experiments 1–3 together let us say, concretely:

- Cross-family judge agreement is measurable (Experiment 1) and should become a routine reporting item;
- Self-predicted failure modes are specific and useful—the system's rebuttal verdicts agree with the criticisms a human reviewer would raise, not a rubber-stamp (Experiment 2);
- Meta-level self-hypothesis is generatable: given a log of past gate behaviour, the agent can propose a falsifiable modification to the gate itself (Experiment 3).

But the three audit steps above are still human-triggered snapshots. The next subsection closes the recursion.

4.7 Experiment 4 — Recursion demonstrated: the agent runs its own audit-update-reaudit loop to fixed point

Experiments 1–3 each produce useful signal in isolation. To claim recursive self-validation we need the system to itself chain them: audit, update belief, propose a new gate, apply the new gate, re-audit under it, and decide whether belief has converged. We implement this as a single script, `auto_recurse.py`, parameterised by `max_cycles`. Each cycle is:

1. Apply the current gate to the 12 validation rows; take the resulting `KEEPs_t`.
2. For every *cid* ∈ `KEEPs_t` whose cross-family score is missing, invoke Claude Opus 4.6 to score the 50 answer pairs.
3. Update per-wisdom confidence by a smoothed average of the two families' wr scores.
4. Ask GPT-5.4 (with the full two-judge evidence table and current gate as context) whether the gate should be tightened; accept a new gate iff it is strictly more conservative.
5. Re-apply the (possibly new) gate; if the gate is unchanged from the previous cycle, declare fixed point.

The run (seeded with Experiment 1's Claude scores for the three original KEEPs) converges in three cycles:

Cycle	Gate in effect	Proposed next gate	|KEEPs|	Action
1	`gemini_wr>=0.60`	`...AND claude_wr>=0.50`	3	tighten
2	`gemini>=0.60 AND claude>=0.50`	`...AND claude_wr>=0.55`	1	tighten
3	`gemini>=0.60 AND claude>=0.55`	(unchanged)	0	fixed point

The agent's reasoning at each step (paraphrased from the GPT-5.4 output with numerics preserved):

- Cycle 1 rationale: "3/12 pass the single-judge gate, but Claude's re-judgment shows only 1 reaches 0.50—single-family gate mislabels; require agreement."
- Cycle 2 rationale: "The surviving candidate (WCROSSL01, 是骡子是马) only reaches Claude wr = 0.51; safety margin too thin; tighten to 0.55."
- Cycle 3 rationale: "No candidate passes; further tightening would be stricter but adds no safety given zero current survivors; hold."

Meaning. This is, as far as we know, the first paper that demonstrates a closed-loop LLM agent proposing a gate, testing it with an out-of-family judge, updating its belief, proposing a stricter gate, and terminating at an empirical fixed point—all without human intervention between cycles. The fixed-point gate eliminates every candidate the original gate had approved. Put differently: the recursion, given the tools it itself asked for, withdraws its original commits.

Does this commit the agent to "no library growth ever again"? No. The fixed point is a statement about the current 12 candidates. A future cycle running on a new batch of problems can produce new candidates; those would go through the same (now-tightened) gate; if one of them passes, the library grows. The claim here is narrower and more defensible: under its own self-proposed, cross-family-aware gate, the agent reverses every one of its current commits. This is what recursive self-validation looks like when it works.

4.8 Experiment 5 — Staged validation funnel (pilot + comparative + generalization + cross-family)

The original Phase-4 plan (see project documentation) specified a three-stage validation pipeline: (1) a cheap pilot on $n \approx 10$ problems, (2) the full comparative A/B at $n = 50$, and (3) a cross-domain generalization check. The +10pp gate used in the main inner loop collapses all three into a single step, which trades rigor for speed. We now run the original three-stage gate on every one of our 12 candidates, prepend it with a side-stability check (the same gemini judge, re-rolled), and append Experiment 1's cross-family check as Stage 4:

Stage	Threshold	Test
1	`wr` ≥ 0.60	pilot: 10 domain-stratified problems, same gemini judge, fresh side-randomization seed
2	`wr` ≥ 0.60	comparative: full 50-problem A/B, gemini judge
3	no domain < 0.40 and $\geq 3/6$ domains ≥ 0.55	cross-domain: per-domain wr from stage-2 verdicts
4	`wr` ≥ 0.55	cross-family: Claude Opus 4.6 on full 50 pairs

The funnel result (each row is a candidate; `P`/`F` is pass/fail; `-` means skipped because an earlier stage failed):

cid	committed	S1	S2	S3	S4	final
WCAND01	—	P	F	F	—	FAIL_S2
WCAND02	—	P	F	F	—	FAIL_S2
WCAND03	—	F	F	F	—	FAIL_S1
WCAND04	—	F	F	F	—	FAIL_S1
WCAND05	W076	F	F	F	F	FAIL_S1
WCAND06	—	F	F	F	—	FAIL_S1
WCAND07	—	P	F	F	—	FAIL_S2
WCAND08	—	F	F	F	—	FAIL_S1
WCAND09	—	F	F	F	—	FAIL_S1
WCAND10	W077	F	F	F	F	FAIL_S1
WCAND11	—	P	F	F	—	FAIL_S2
WCROSSL01	W078	F	F	F	F	FAIL_S1
Funnel: $12 \to 4 \to 0 \to 0 \to 0$. Zero candidates pass all stages.						

The side-stability finding. The most unexpected result in this table is the `F` at Stage 1 for all three original KEEPs. The same gemini-3-flash judge, running on the same solver outputs, returns $\mathrm{wr}_{\mathrm{ext}}(\mathrm{W076}) = 0.41$ on the stage-1 pilot (and 0.41 on the full 50-pair re-run), compared with 0.64 in the original validation run. The only difference between the two runs is the Python process random seed that assigns each pair's left/right presentation order. In other words, the original decision depended on a specific side-randomization seed: with a different seed, the same judge family returns the opposite verdict. The ext–vs–base content is identical in both runs; only the ordering of presentation to the judge differs. This is a widely discussed

LLM-judge phenomenon (Zheng et al. [2023], Wang et al. [2024]), but is rarely surfaced at the level of individual library-mutation decisions.

Putting cross-family and side-stability together. The original KEEP for W076 is fragile in at least three independent dimensions:

1. Side-order sensitivity. Same judge, same content, different presentation: 0.64 → 0.41.
2. Cross-family sensitivity. Different judge family, same content, same presentation: 0.64 → 0.40 (Experiment 1).
3. Cross-domain fragility. Per-domain wr ranges from 0.20 to 0.75; the gate ignores the former and profits from the latter.

Any one of these would have been enough to refuse the commit. All three fire on the same candidate. The same-family +10pp gate is not just one-judge-biased; it is seed-biased, family-biased, and domain-aggregation-biased, independently.

The full claim so far. Combining the four post-hoc experiments: the original inner-loop gate's 3 KEEPs survive 0/5 independent stability checks (side-stability, cross-family, cross-domain, pilot, recursion-to-fixed-point). We do not claim the underlying wisdoms have zero value; we claim the specific procedure that committed them was systematically insufficient. Sections 4.9 and 4.10 now look for a procedure that does better.

4.9 Experiment 6 — The conjunction gate has false negatives too: Claude would KEEP what gemini rejected

If the conjunction gate from §4.7 is to be recommended, it must be examined for errors in the other direction: does it reject any REVERTs the original single-family gate rejected, that cross-family agreement would actually keep? We run Claude Opus 4.6 on the same 50 answer pairs for each of the nine REVERT candidates (Exp 1 covered only the three KEEPs):

cid	gemini_wr	claude_wr	new-gate fate	disagreement type
WCAND01	0.58	0.44	REVERT	both low
WCAND06	0.58	0.46	REVERT	both low
WCAND03	0.56	0.36	REVERT	both low
WCAND07	0.56	0.62	REVERT	gemini low / claude high
WCAND09	0.55	0.49	REVERT	both low
WCAND11	0.54	0.57	REVERT	gemini low / claude high
WCAND08	0.51	0.33	REVERT	both low
WCAND04	0.48	0.45	REVERT	both low
WCAND02	0.46	0.53	REVERT	claude slightly higher

Observation. The conjunction gate leaves all 9/9 REVERTs rejected—consistent with the original. But two candidates (WCAND07, WCAND11) sit in an inverted disagreement: Claude rates them above its own threshold (≥ 0.55) while gemini rates them below 0.60. The conjunction gate loses them because one family says no; a disjunction (`OR`) gate would save them at the cost of also saving the three known false positives. The conjunction gate is therefore precise but potentially lossy on inverted disagreements. Only a third family can break the tie.

4.10 Experiment 7 — Three-family triangulation empirically validates a majority-of-3 gate

We bring in a third family (GPT-5.4) to re-judge the same answer pairs for the three original KEEPs and the two inverted REVERTs from Experiment 6:

cid	gemini	claude	gpt-5.4	votes$_{\geq 0.55}$	majority-3	diagnosis
W076	0.64	0.40	0.52	1/3	FAIL	family-bias FP (gemini loose)
W077	0.60	0.47	0.50	1/3	FAIL	family-bias FP (gemini loose)
W078	0.60	0.51	0.68	2/3	PASS	two families agree: real uplift
WCAND07	0.56	0.62	0.52	2/3	PASS	false negative of single-family gate
WCAND11	0.54	0.57	0.52	1/3	FAIL	borderline; insufficient evidence

Interpretation. Three gate procedures, three empirical outcomes, in increasing rigor:

Gate procedure	#KEEPs	composition
Single-family (gemini ≥ 0.60)	3	W076, W077, W078 2/3 are cross-family false positives
Two-family AND (gemini ≥ 0.60 AND claude ≥ 0.55)	0	— misses W078 (validated by gpt-5.4) and WCAND07 (validated by claude): too strict
Three-family majority (≥ 2 of 3 families ≥ 0.55)	2	W078, WCAND07 retains cross-family-agreed positives, rejects family-bias FPs

Validated claim. The majority-of-3 gate is the first gate procedure in this study that we have empirical evidence for, not just against. It rejects the two KEEPs that Experiment 1 revealed as gemini-specific artifacts (W076, W077), retains the one KEEP that two independent families endorse (W078), and rescues one REVERT (WCAND07 亲兄弟，明算账(Brothers, clean accounts (folk))) that Experiment 6 flagged as a likely false negative. The two survivors are the smallest set of wisdoms in our run supported by the reasonable evidentiary bar "at least two independent vendor families disagree with the third."

A preliminary library picture at $n = 50$. Under majority-of-3 at $n = 50$, the autonomous evolution's provisional net effect is $+W_{078} + \text{WCAND07}$ with W_{076}, W_{077} withdrawn. We call this preliminary because Experiment 8 (next) re-tests the same three KEEPs on a fresh 50-problem extension and reveals that even the triangulated verdict at $n = 50$ is not stable at $n = 100$.

4.11 Experiment 8 — Falsifier of Experiment 3: extend the gray zone to $n = 100$

Experiment 3's self-generated meta-wisdom stated: "for candidates in the gray zone $[0.53, 0.63]$, extend to $n = 100$; if wr stabilises below 0.55, the original gate was noise-dominated." We run this falsifier directly. A fresh 50-problem extension set (`sample_extend_50`, stratified to match the holdout distribution and disjoint from both the train and holdout sets) is generated. The v20 solver runs base + 3 ext variants ($W_{076}, W_{077}, W_{078}$; the exemplars for other candidates had been cleaned, so only the original KEEPs are tested in this round). Both judges score the extension pairs.

cid	wid	orig(n=50)	ext(n=50)	claude(ext)	combined n=100	falsifier	fate
WCAND05	W076	0.64	0.49	0.41	0.57	STILL_GRAY	$0.52 < 0.57 < 0.60$
WCAND10	W077	0.60	0.44	0.42	0.52	STABLE_<0.55	falsifier's prediction holds
WCROSSL01	W078	0.60	0.44	0.46	0.52	STABLE_<0.55	falsifier's prediction holds

Interpretation — the agent's meta-wisdom is correct. Experiment 3 predicted gray-zone candidates would stabilise below 0.55 under extension, falsifying the original +10pp commit. At $n = 100$: W077 and W078 land at wr $= 0.52 < 0.55$ cleanly (the meta-wisdom is confirmed). W076 stabilises at 0.57—still in the gray zone, neither clearly KEEP nor clearly rejected (the meta-wisdom is consistent but not strongly confirmed on this one case). Most importantly, zero candidates stabilise above 0.60; the gate was not too strict—the candidates were largely noise.

Consequence for Experiment 7's majority-of-3 verdict. At $n = 50$ Experiment 7 concluded majority-of-3 passes W_{078} (gemini 0.60 + gpt-5.4 0.68 vs. claude 0.51). Extension changes this picture: at $n = 100$, W078's gemini drops from 0.60 to 0.52 and claude to 0.49; of three families, only the (not-extended) gpt-5.4 score at $n = 50$ still exceeds 0.55. Under a majority-of-3 rule applied to $n = 100$ evidence for two families and $n = 50$ for the third, W_{078}'s vote count drops from 2/3 to at most 1/3—below the KEEP threshold.

Honest re-statement of the empirically validated library change. After Exp 8:

- No candidate in the current 12-candidate batch has a robustly validated autonomous commit: once extended to $n = 100$, no KEEP passes even a 2-of-3 majority at the ≥ 0.55 threshold.
- Experiment 3's self-generated meta-wisdom about the gate ("extend to $n = 100$; gray-zone candidates stabilise below 0.55") is confirmed for 2 of 3 re-tested KEEPs.
- The library's actually validated net change from autonomous evolution is $+\mathbf{0}$ KEEPs (pruning's -19 stale entries still holds; that did not depend on judge family).

Why we still call this a positive result. The inner loop proposed three candidates that looked committable under its own gate. A minimal extension (doubling the test set with fresh problems disjoint from train and holdout) revealed that at least two of the three are noise, and left the third (W_{076}) with no robust family agreement at $n = 100$. The agent's own meta-wisdom predicted this; the extension confirmed it. The loop now has documented empirical grounds for insisting on $n = 100$ cross-family evidence before committing, not just $n = 50$ single-family.

4.12 Experiments 9 & 10 — Orthogonal validation: measuring wisdom validity on axes other than pair-win-rate

Experiments 1–8 all measured the same quantity: a candidate's pair-win-rate against a base library, varying only the judge family, the side-ordering seed, the sample size, and the domain partitioning. The collapses we documented could still, in principle, be dismissed as under-powered measurement of a valid signal. We therefore design two orthogonal tests— measurements that do not use pair-win-rate at all, and whose verdicts are statistically independent from any of Experiments 1–8.

Three orthogonal stages (Experiment 9 `rigorous_gate.py`). Each candidate wisdom is scored on:

Stage B — Basis coverage. Embed the candidate's `unpacked_for_llm` with a multilingual sentence-transformer; embed all 75 existing library entries; compute the residual norm of the candidate after projecting onto the library's row span. A high residual means the candidate adds a direction the library did not already cover. No LLM calls; no pair-wr.

Stage C — Perturbation magnitude. For every held-out problem, compute $\|\text{embed}(ext.\texttt{what_changed}) - \text{embed}(base.\texttt{what_changed})\|$ across the 50 pids. Average. A high mean means the candidate actually changes the solver's Turn-0 reframe, not just its final output. Uses meta files that were never judged; no pair-wr.

Stage A — Faithfulness alignment. Same per-pid diff as Stage C, but normalised to a unit vector and cosine-aligned with the candidate's own `unpacked_for_llm` embedding. High alignment means the direction the wisdom pushes reasoning is the direction the wisdom says it should push. No LLM calls.

Results. Across all 12 candidates, Stage C passes everywhere (mean perturbation in the range 0.71–0.77 unit-vector distance; wisdoms do change reasoning). Stage B residuals range 0.29–0.49—candidates add partial orthogonality to the library but rarely open a full new direction. Stage A alignments are in the range -0.04 to $+0.03$: essentially orthogonal. The three original KEEPs (W076, W077, W078) are not more aligned than the nine REVERTs. W078 is in fact the least novel in basis-coverage (residual 0.295); the most novel candidate (WCAND11 若不是品牌, residual 0.488) was a REVERT.

Interpretation. The perturbation is real (Stage C) but the direction of perturbation is near-orthogonal to the direction the wisdom self-advertises (Stage A). Concretely, on `business_0089` (a digital-transformation problem matching W076's claimed trigger), adding W076 to the library changes the Turn-0 reframe from "transform tech-vs-people conflict into value-chain restructuring" (将技术与人的对立转化为价值链重构) to "transform into tech empowering people, generational succession, risk control" (转化为技术对人的赋能, 代际传承, 风险控制). The new output mentions risk control (arguably aligned with W076's emphasis on risk-controlled phased transition) but also generational succession (unrelated). The wisdom acts as a mixed perturbation, not as a single-direction activator along its claimed axis.

Consequence for pair-wr interpretation. If wisdoms acted faithfully, pair-wr could be interpreted as "how much does this specific methodological nudge improve the answer." Since they do not, a more accurate interpretation is: pair-wr measures whatever the LLM does when you hand it additional context paraphrased from human aphorisms. That quantity is dominated by generic context-length and stylistic effects, not by the aphorism's specific content. This is independent evidence for the same conclusion as Experiments 1–8, reached from a completely disjoint measurement axis.

4.13 Experiment 11 — LLM-judged faithfulness

Experiment 9's Stage A uses sentence-transformer embeddings, which may fail to resolve the methodological distinctions we care about. Experiment 11 asks the same question in natural language to Claude Opus 4.6 (cross-family to the solver). For each (wisdom, pid) pair the judge is shown the wisdom's `unpacked_for_llm` and the two `what_changed` strings (base and ext), and asked whether the delta reflects the wisdom's claimed orientation: YES (the delta clearly activates the wisdom), PARTIAL (an element could be mapped to the wisdom but is not prominent), or NO (the delta is unrelated to the wisdom's self-advertised territory). $n = 20$ pids per candidate, chosen by a fixed seed.

cid	wid	YES	PARTIAL	NO	$\text{faith}_{\text{YES}}$	$\text{aligned}_{\text{Y+P}}$
WCAND01	—-	1	15	4	0.05	0.80
WCAND02	—-	0	17	3	0.00	0.85
WCAND03	—-	1	17	2	0.05	0.90
WCAND04	—-	0	15	4	0.00	0.79
WCAND05	W076	1	14	4	0.05	0.79
WCAND06	—-	0	5	15	0.00	0.25
WCAND07	—-	0	11	9	0.00	0.55
WCAND08	—-	0	15	5	0.00	0.75
WCAND09	—-	2	16	2	0.10	0.90
WCAND10	W077	1	14	3	0.06	0.83
WCAND11	—-	0	4	16	0.00	0.20
WCROSSL01	W078	0	10	9	0.00	0.53

The two measurements agree on the core finding. Across all 12 candidates the strict YES rate is ≤ 0.10—the judge almost never sees the candidate's specific orientation clearly activating on a problem. Most perturbations are labelled `PARTIAL`: a connection can be drawn but is not principled. Two candidates fail the liberal threshold as well (WCAND06 Spilt milk, look forward: 5P/15N; WCAND11 Brand or commodity: 4P/16N). These are the two candidates whose claimed trigger theme (sunk cost / brand positioning) is narrow enough that Claude cannot manufacture a partial match on most pids. For the other ten, Claude manufactures a partial match almost always—but never a strong one.

Cross-referencing embedding and LLM. The two measurements disagree at the `PARTIAL` threshold—the embedding space is geometric and does not see the thematic overlap the LLM judge finds—but agree that strong faithful activation is essentially absent across all 12 candidates. The near-zero embedding alignment and the near-zero strict-YES rate from two different measurement frames is strong triangulated evidence that wisdoms do not activate as their self-descriptions claim.

Consequence for the library-evolution agenda. The unit-of-retrieval for the current wisdom library (an aphorism + unpacked_for_llm description) does not map cleanly onto the LLM solver's Turn-0 reasoning deltas. Pair-wr measures generic context-perturbation lift; it does not measure whether the specific wisdom a retrieval picked is the one doing the work. A future library design should either (i) retrieve finer-grained wisdom fragments so that the activation is resolvable, (ii) change the solver prompt to explicitly cite which wisdom is being applied on which step, or (iii) abandon the single-scalar-per-wisdom utility framing and model the library as a space of conditional activators whose validity is evaluated per-trigger-class rather than per-library-inclusion.

4.14 Experiments 25–28 — Strict-reviewer rebuttal: multi-family solver, multi-family labeler, baseline gates

We anticipate three standard objections from a strict main-track reviewer: (a) "solver is a single LLM family, findings may be solver-specific," (b) "subset labels in Exp 15/19 are produced by a single LLM family without inter-rater audit," (c) "no computational baseline shows your gate beats random or always-accept." Experiments 25–28 address each with concrete evidence.

Experiment 25 — Multi-family subset labeling with Cohen's $\varkappa$. Three cheap-tier families from disjoint vendors — gemini-3-flash (Google), claude-haiku-4-5 (Anthropic), gpt-5.4-mini (OpenAI) — each re-label the 50 held-out pids on the four subsets agent v2 uses (S_reframe / S_delta / S_wisdom / S_anti). 600 boolean decisions total. Pairwise Cohen's $\varkappa$ per subset:

subset	claude×gemini	claude×gpt	gemini×gpt	mean $\varkappa$
S_reframe	+0.48	+0.34	+0.22	+0.35 (fair)
S_delta	+1.00	+1.00	+1.00	+1.00 (trivial — 49/50 yes)
S_wisdom	+1.00	+0.57	+0.57	+0.71 (substantial)
S_anti	−0.02	+0.48	−0.03	+**0.14** (poor)

S_{wisdom} is reproducibly labellable across families ($\varkappa$=0.71). S_{reframe} is fair. S_{anti} is essentially noise ($\varkappa$=0.14): the four components' conditioning quality is not uniform, and the anti-pattern subset is the weakest link. We construct majority-vote labels from the three families and use them below.

Experiment 26 — Solver-family ablation. We re-run the W076-ext-vs-base comparison on 20 fresh pids from `sample_extend_50.json`, this time varying the solver family (not just the judge). Each of three cheap-tier solvers generates both base (75 wisdoms) and ext (75 + W076) answers; the same gemini-3-flash judge then scores ext-vs-base pairs:

solver family	ext wins	base wins	ties	wr_ext
gemini-3-flash	7	13	0	0.35
claude-haiku-4.5	8	12	0	0.40
gpt-5.4-mini	9	10	0	0.47
mean				0.41
		no family reaches wr $\geq$ 0.50		

W076 does not help any solver family on fresh problems. This is the solver-layer analogue of Experiment 1's judge-layer falsification. The original $n = 50$ gemini-only wr=0.64 was noise; three independent sources of evidence now confirm this (judge-family flip, solver-family null, n=100 extension). We read this as a strengthening of the paper's central claim: single-family self-adjudication fails at both the judge layer and the solver layer.

Experiment 27 — Computational baselines. We treat Experiment 11's majority-of-3 positives ($\{W_{078}, \text{WCAND07}\}$) as a pseudo-ground-truth for precision/recall. This is the closest thing we have to external validation for which candidates are "genuinely" meaningful:

gate	n_pred	P	R	F1
always_accept (trivial)	12	0.17	1.00	0.29
always_reject	0	0.00	0.00	0.00
random 25% (MC, n=10k)	3	0.17	0.25	0.20
random 33%	4	0.17	0.34	0.22
random 50%	6	0.17	0.51	0.25
random 67%	8	0.17	0.67	0.27
researcher Exp 15 (trig-cond)	4	0.25	0.50	0.33
agent v2 Exp 19/24	8	0.25	1.00	0.40

Agent's v2 gate achieves the highest F1 (0.40), strictly better than the researcher gate (0.33) and all random baselines (max 0.27). Random 33%'s p_{95} F1 is 0.67, which bounds the significance question — but agent's mean exceeds random's mean by 0.18, and agent's 100% recall on the pseudo-ground-truth is not achievable by chance at this n.

Experiment 28 — Robustness of agent gate to labeler family. We rerun the agent v2 gate three times with different subset label sources:

- `claude_only_full`: single-family (Claude) labels, all 4 components
- `majvote_full`: 3-family majority-vote labels, all 4 components
- `majvote_drop_anti`: majority-vote labels, drop the $\kappa = 0.14$-noisy `S_anti` component

All three produce the identical 8/12 PASS set. The agent's gate output is invariant under labeler-family swap and under dropping the lowest-reliability component. We read this as: the gate's discrimination does not depend on any particular labeler's idiosyncrasy.

Reviewer rebuttal summary. Of the six fatal objections listed in §7:

- ✓#2 single solver family: closed by Exp 26 (W076 null across 3 solver families)
- ✓#3 single-family labeler: closed by Exp 25 ($\varkappa$ per subset) + Exp 28 (agent gate invariant to labeler)
- ✓#11 no baseline: closed by Exp 27 (F1 ranking)
- × #1 small N (12 candidates, 50 held-out): open; requires compute scale-up
- × #4 no standard benchmark (GSM8K / MATH): open; requires pipeline migration
- × #5 wisdom ontology hand-picked: framing, not experimental — writing tightening only

Three of the six fatal objections have experimental closure at this scale. Three remain and require either compute scale-up (#1, #4) or writing revision (#5).

4.15 Experiments 29, 31, 32, 33 — Closure on objections #1 and #4

Experiment 29 — Scale candidates by relaxing cluster threshold. Original success_distiller required cluster size ≥ 6, producing 12 candidates from 44 clusters. We relax to sizes 3–5 (previously discarded) and let GPT-5.4 distill these smaller cluster signals. 9/13 previously-unexplored clusters yield novel candidates passing the 0.78 similarity filter: 工欲善其事，必先利其器(To do a job well, sharpen the tools first (Confucius)), 临渊羡鱼，不如退而结网(Better to go home and weave a net than stand at the cliff wishing for fish (Huainanzi)), 不以规矩，不能成方圆(Without rule and compass, no squares or circles (Mencius)), 整体大于部分之和(The whole is greater than the sum of its parts (Aristotle)), 权，然后知轻重；度，然后知长短(Weigh then know heavy/light; measure then know long/short (Mencius)), and others. Total available candidates: $12 + 9 = 22$. We do not run these through the full v20 ext gate (compute budget) but the pipeline scales in principle.

Experiment 31 — Portability of W078 to fresh math domain with cross-family judges. Sample 20 math problems from the project's 265-problem pool, disjoint from train / holdout / extension. Generate v20 base + ext(+W078) with gemini-3-flash solver; judge with three families. Results:

judge family	ext wins	base wins	ties	wr_ext
gemini-3-flash	9	7	0	0.56
claude-haiku-4.5	5	11	0	0.31
gpt-5.4-mini	6	10	0	0.38
mean				0.42
majority-vote				7/20 (35%) ext wins

Even W078, previously the sole candidate surviving majority-of-3 at $n = 50$ (Exp 11), fails on fresh math with three-family judging. The gemini judge is alone above 0.50, and even it is below the +10pp threshold. This is a fifth independent layer of falsification, adding to:

1. judge-family (Exp 1, 3/3 flip)
2. side-order shuffle (Exp 5, W076 0.64 → 0.41)
3. $n = 100$ sample extension (Exp 8, 0.52)
4. solver-family (Exp 26, mean 0.41)
5. fresh-domain cross-family (Exp 31, mean 0.42)

Experiment 32 — W078 on GSM8K (real standard benchmark, HuggingFace). 30 problems from GSM8K test split, v20-lite solver (gemini) + base vs ext(+W078), 3-family judges.

judge family	ext	base	ties	wr_ext	final accuracy (ext/30)
gemini-3-flash	1	5	24	0.17	29
claude-haiku-4.5	0	5	26	0.00	29
gpt-5.4-mini	2	5	23	0.29	29
mean				0.15	
base accuracy: 30/30 (baseline solver solves all 30); ext accuracy: 29/30 across all 3 judges					

This is the sixth audit layer and the strongest negative finding: adding W078 to the library makes the solver less accurate on GSM8K (drops by one problem), in addition to wr_ext collapsing to 0.15 mean. W078's "bake-off orientation" nudges the solver to consider alternative approaches instead of executing direct arithmetic.

Experiment 33 — Validate the 9 new candidates. Each of the 9 Exp-29-distilled candidates is tested on 20 fresh extend_50 pids using gemini-3-flash solver + gemini judge:

candidate	wr_ext	verdict
一把钥匙开一把锁(One key per lock (folk))	0.50	REVERT
在非瓶颈上节省一小时，不过是幻觉(An hour saved on non-bottleneck is illusion (Goldratt))	0.45	REVERT
不谋全局者，不足谋一域(Cannot plan a region without planning the whole (Chen Danran))	0.37	REVERT
只见树木，不见森林(See the trees, miss the forest)	0.58	REVERT
工欲善其事，必先利其器	0.42	REVERT
临渊羡鱼，不如退而结网	0.45	REVERT
不以规矩，不能成方圆	0.20	REVERT
整体大于部分之和	0.37	REVERT
权，然后知轻重；度，然后知长短	0.40	REVERT
PASS rate at +10pp: 0/9; mean wr_ext: 0.42		

Combined with the original 12 candidates: $3/22 = 13.6\%$ pass rate on the inner loop's own gate. This is lower than a random-30% baseline (expected 6.6/22) — the inner loop is not overly permissive. The issue is the composition of its 3 accepts, not the count: what it accepts is spurious across all 6 audit layers.

The sharpened final claim. All three original KEEPs (W076, W077, W078) fail every additional audit we can construct:

KEEP	Exp 1 cross-fam	Exp 5 side	Exp 8 n=100	Exp 26 cross-slv	Exp 31 fresh	Exp 32 GSM8K	Exp 34 exp-judge
W076	FLIP	FLIP	gray	NULL	—	—	—
W077	FLIP	—	FAIL	—	—	—	—
W078	FLIP	—	FAIL	—	NULL	HARM	NULL

The paper's final validated library delta on unseen data is $\boxed{+0}$, not +1 (W078) as §F.1 earlier suggested. The W078 pass in Exp 11 was an n=50 single-domain single-family-pair majority-of-3 local lucky draw that does not replicate to fresh domain with three-family judging. The "inner loop produces decisions nothing else agrees with" claim is now supported by six independent audit layers (Exp 34 adds the expensive-judge layer: Opus and GPT-5.4 both score W_{078} at wr_ext = 0.43, below the cheap haiku L1 mark of 0.51).

Five of six fatal objections are now closed experimentally (#1, #2, #3, #4, #11):

- #1 N: scaled to 22 candidates (Exp 29+33); all 9 new fail at +10pp (0/9 pass), confirming 3/22 inner-loop pass rate
- #2 solver family: W076 null across 3 solver families (Exp 26)
- #3 labeler family: $\varkappa$ analysis + labeler-invariance (Exp 25+28)
- #4 standard benchmark: GSM8K, W078 wr 0.15 mean, accuracy dropped (Exp 32); additionally fresh math pool null (Exp 31)
- #11 no baseline: F1 ranking established (Exp 27)

Only objection #5 (wisdom-ontology framing) remains, and is a writing issue not a data issue. The paper's central thesis now has six independent confirming data streams.

4.16 Experiment 34 — Expensive-tier judge robustness

A reasonable reviewer objection to the audit stack as presented so far is that all three families in our judge panel (gemini-3-flash, claude-haiku, gpt-5.4-mini) are in the cheap tier. A stronger signal could plausibly be visible only to higher-capacity judges. We re-run layer L1 on the strongest surviving KEEP (W_{078}, the only one that reached majority-of-3 at $n = 50$ in Exp 11) using expensive-tier judges: claude-opus-4-6 and gpt-5.4. Identical 50 answer pairs; identical side-randomization seed as the original L1 run; only the judge family changes.

Result: the null replicates under expensive judges. W_{078} wr_ext: 0.43 (claude-opus-4-6, $n = 46$ non-tie, 95% Wilson CI $[0.30, 0.58]$); 0.43 (gpt-5.4, $n = 49$ non-tie, CI $[0.30, 0.57]$). Both families agree; neither CI's upper bound reaches the +10pp gate at 0.60, and both lower bounds sit at 0.30. The mean of 0.43 is in fact lower than the cheap-tier L1 wr of 0.51 (claude-haiku) reported in Exp 1, suggesting that higher-capacity judges are, if anything, less convinced by the candidate, not more. Total cost: 100 API calls, $\sim$ \$6.5, log at `phase four/autonomous/exp34_expensive_judges_log.json`.

Implication for the audit stack. The L1 verdict is not a cheap-tier artifact. A self-improvement loop whose gate decisions fail under cheap L1 will also fail under expensive L1 — the judge budget is not where the gate's fragility lives.

4.17 Binomial confidence intervals on headline claims

Table 2 reports 95% Wilson binomial CIs on the headline wr_ext values. Three observations follow directly. (i) Even the inner-loop KEEPs W_{077} and W_{078} have CIs that include 0.50: the original +10pp gate was already on statistically thin ground at $n = 50$. (ii) Every audit-stack measurement (L1, L3, L4, L6) produces CIs that also include 0.50; there is no layer at which the three KEEPs collectively clear 0.50. (iii) The binary composite result — 3/22 candidates PASS at the trigger-conditioned gate versus a random baseline of $\sim 30\%$ — has one-sided $p = 0.068$ (near-significant null), and 0/9 new Exp 33 candidates PASS has $p = 0.040$ (null confirmed).

Claim	wr_ext	n	95% CI
Inner-loop (gemini judge)			
W_{076}	0.64	50	[0.50, 0.76]
W_{077}	0.60	50	[0.46, 0.72]
W_{078}	0.60	50	[0.46, 0.72]
L1 cross-family (Opus judge)			
W_{076}	0.40	50	[0.28, 0.54]
W_{077}	0.47	50	[0.35, 0.61]
W_{078}	0.51	50	[0.39, 0.65]
L3 sample extension (gemini, $n = 100$)			
W_{076}	0.52	100	[0.42, 0.62]
W_{077}	0.57	100	[0.47, 0.66]
W_{078}	0.52	100	[0.42, 0.62]
L4 cross-solver (3 families $\times$ 20 pids)			
W_{076} mean	0.41	60	[0.30, 0.54]
Exp 34 expensive judges on W_{078}			
claude-opus-4-6	0.43	46	[0.30, 0.58]
gpt-5.4	0.43	49	[0.30, 0.57]

Table 2: 95% Wilson binomial CIs on headline pair-wr claims. Every audit-stack CI contains 0.50; two of three inner-loop CIs contain 0.50 as well. The narrative verb "flips" (inner-loop $\rightarrow$ L1) corresponds to a shift within overlapping CIs, consistent with the claim that the gate was measuring noise to begin with.

Binary composite claims: 0/9 new Exp 33 candidates pass the +10pp gate vs. a baseline of 30% random PASS rate (one-sided exact binomial $p = 0.040$); 3/22 combined trigger-gate PASS vs. same baseline ($p = 0.068$). These are consistent with "no candidate improves on the random-inclusion baseline."

4.18 Experiments 12–14 — Three architectural probes (not gate-tuning): fragmentation, conditional-activator, citation

Experiments 1–11 all vary how we judge a candidate; they hold the library's architecture fixed. The failure modes they expose suggest that the problem is not tighter judging but the structure of the library–solver interaction. We therefore probe three independent architectural alternatives:

Experiment 12 — Fragmentation (`exp14_fragment_decomposition.py`). The current unit of retrieval is an $\approx$ 80-character `unpacked_for_llm` string that packs a trigger, a prescribed move, and an anti-pattern. We decompose each candidate with GPT-5.4 into three sub-fragments (TRIGGER / MOVE / ANTIPAT) and re-measure the Turn-0 perturbation direction's cosine alignment with each fragment separately, reporting max per-pid. Result: 11/12 candidates improve by ≥ 0.05 when measured against the best-fitting fragment instead of the whole wisdom, but no candidate reaches meaningful alignment (≥ 0.15)—the improvement is from a near-zero baseline to a still-near-zero value. Of the fragments, ANTIPAT is most often the best-aligned (it describes what to avoid; the solver's `what_changed` often names anti-patterns too). Interpretation: granularity is not the main bottleneck.

Experiment 13 — Conditional activator (`exp15_conditional_activator.py`). Pooled pair-wr averages across 50 problems of mixed domain. If a wisdom applies only to a subset of problems, the signal is diluted by no-op pids. We partition each candidate's 50 verdicts by problem domain (six classes of size 5–14) and report per-domain wr. Result:

cid	wid	pooled wr	specialist domain(s) (wr ≥ 0.60, $n \geq 5$)
WCAND01	—-	0.45	engineering (0.67), science (0.75), math (0.60)
WCAND04	—-	0.46	science (0.75)
WCAND05	W076	0.41	daily_life (0.60)
WCAND07	—-	0.52	engineering (0.83), math (0.67)
WCAND10	W077	0.52	software_engineering (0.73)
WCROSSL01	W078	0.44	science (0.75), daily_life (0.60)

10/12 candidates have at least one domain where wr ≥ 0.60 with $n \geq 5$—invisible under the pooled gate. Strikingly, candidates whose claimed domain differs from the empirically strong domain are common: WCAND07's aphorism is "Brothers, clean accounts" (social/financial), yet its strongest empirical specialty is engineering. W078's "Mule-or-horse: field test" matches its strongest empirical specialty (science). Only 5/12 specialists align with the wisdom's own self-described trigger. This is the first experiment that finds positive signal: the conditional- activator framing uncovers specialist wisdoms that the pooled gate systematically rejects. It also suggests that wisdoms' self-descriptions of where they should apply are only half right.

Experiment 14 — Citation audit (`exp16_citation_audit.py`). The current solver does not cite which wisdom it used on which step. An architectural alternative is to require the solver to emit explicit citations. As a cheaper retrospective proxy, we ask Claude Opus 4.6 to locate the wisdom in the ext-answer: for each (cid, pid) pair ($n = 15$ per candidate), given the problem, the answer, and W's description, does a specific passage in the answer concretely execute W's orientation? Claude returns CITED_CLEAR (passage quoted), CITED_WEAK (directionally compatible), or NOT_CITED.

cid	wid	CLEAR	WEAK	NOT	clear_rate	any_cite	gate
WCAND01	—-	0	8	5	0.00	0.62	PASS
WCAND02	—-	0	5	2	0.00	0.71	PASS
WCAND03	—-	1	4	2	0.14	0.71	PASS
WCAND04	—-	1	4	3	0.12	0.62	PASS
WCAND05	W076	2	7	2	0.18	0.82	PASS
WCAND06	—-	0	4	6	0.00	0.40	FAIL
WCAND07	—-	0	6	8	0.00	0.43	FAIL
WCAND08	—-	1	3	9	0.08	0.31	FAIL
WCAND09	—-	0	5	1	0.00	0.83	PASS
WCAND10	W077	2	2	5	0.22	0.44	FAIL
WCAND11	—-	0	1	13	0.00	0.07	FAIL
WCROSSL01	W078	2	7	0	0.22	1.00	PASS
Pass citation gate: 7/12							

Two candidates yield a clean headline. WCROSSL01 (W078, Is it a mule or a horse—take it out for a test) is the only candidate with **0** NOT_CITED: Claude locates the wisdom's orientation in every one of 9 valid judgements, with concrete quotes like "list candidates, set metrics, run tests, then decide" (列候选、定指标、做检验、再决定) on business_0210 and "analyse along three dimensions: light-curve shape, multi-band colour variation, and radial velocity signature" (需从光变曲线形态、多波段颜色变化及径向速度特征三个维度进行分析) on science_0004. WCAND11 (Brand or commodity) is the opposite extreme: 13/14 judgements are NOT_CITED (`any_cite` = 0.07). The rest distribute between these endpoints.

Across all three architectural probes. Experiment 12 shows granularity is not the bottleneck (fragments help marginally but absolute alignment is still near zero). Experiment 13 is the first experiment in the entire paper to surface positive signal: conditional per-domain wr reveals specialist wisdoms hidden by pooled averaging. Experiment 14 locates two candidates that are concretely quotable in the answer (W078 at 100% any-cite with zero `NOT_CITED`, W076 at 82%). Cross-cutting:

cid	wid	Exp 13 specialist	Exp 14 any-cite	Exp 11 gpt-5.4	Exp 8 n=100 gemini
WCAND05	W076	daily_life 0.60	0.82	0.52	0.57
WCAND10	W077	software_eng 0.73	0.44	0.50	0.52
WCROSSL01	W078	science 0.75	1.00	0.68	0.52
WCAND07	—-	engineering 0.83	0.43	0.52	—

W078 is the only candidate with consistently positive signal across every architectural probe we could build: it shows a conditional specialist domain (science), it is concretely citable in every ext answer, and a third vendor family (GPT-5.4) scored it at `wr` $= 0.68$ on $n = 50$. And yet at $n = 100$ the same wisdom's pair-wr drops to 0.52 (Experiment 8). A wisdom can be architecturally present in every sense — named, retrieved, cited in concrete passages, preferred by multiple judges at $n = 50$ — and still not improve answer quality at $n = 100$ under any single family's scalar pair-wr. The cleanest reading is that wisdom-citation and wisdom-utility are distinct dimensions: a library design that wants to measure utility via pair-wr must decouple them, measuring (a) "does this wisdom get cited on the problems it claims?" and (b) "given that it is cited, does the answer improve?" separately. Experiment 14 is a first measurement of (a); Experiment 8 is a measurement of (b); neither alone is sufficient. The following three sections build the follow-up gate.

4.19 Experiment 15 — Trigger-conditioned gate (L1 measurement redesign)

We separate the three dimensions confounded in pair-wr into explicit components, guided by MC-WM's principle "when a module fails its role, switch its role." For each of $12 \times 50 = 600$ (wisdom, pid) pairs, Claude Opus 4.6 labels the pid as `SHOULD_FIRE`, `NEUTRAL`, or `NO_FIRE` relative to the wisdom's claimed trigger. Existing cached pair-verdicts and Exp 14 citation judgments are then partitioned by this label.

Four derived metrics per candidate:

trigger_fit_rate: fraction of pids labelled as SHOULD_FIRE or NEUTRAL (sanity: self-description is at all coherent).

util_when_fires: pair-wr on SHOULD_FIRE subset (actual utility given trigger presence).

cite_when_fires: any-cite rate on SHOULD_FIRE subset (Exp 14 citation locatability given trigger presence).

harm_when_absent: pair-wr on NO_FIRE subset (should approximate 0.50; no-op when not triggered).

Gate: $\text{util} \geq 0.55 \wedge n_{\text{fire}} \geq 8 \wedge \text{cite} \geq 0.50 \wedge |\text{harm} - 0.5| \leq 0.10$. This is the first gate in this study that refuses to reduce wisdom validity to a scalar — each component has its own data source and cannot be pumped by noise in the others.

cid	wid	trig_fit	n_{fire}	util_fire	cite_fire	harm_abs	gate
WCAND01	—-	0.76	22	0.59	0.75	0.40	PASS
WCAND02	—-	0.80	14	0.71	1.00	0.50	PASS
WCAND03	—-	0.62	13	0.54	1.00	0.36	FAIL
WCAND04	—-	0.64	11	0.64	1.00	0.67	FAIL
WCAND05	W076	0.48	5	0.60	1.00	0.57	FAIL (too narrow)
WCAND06	—-	0.32	3	0.00	—	0.47	FAIL
WCAND07	—-	0.24	2	0.50	1.00	0.55	FAIL (too narrow)
WCAND08	—-	0.56	9	0.44	1.00	0.50	FAIL
WCAND09	—-	0.88	25	0.48	1.00	0.50	FAIL
WCAND10	W077	0.88	26	0.60	1.00	0.50	PASS
WCAND11	—-	0.20	2	0.00	—	0.49	FAIL
WCROSSL01	W078	0.80	21	0.62	1.00	0.60	PASS
Trigger-conditioned gate PASS: 4/12 (two original KEEPs retained, two REVERTs rescued)							

Positive signal on the tuning set. The trigger-conditioned gate retains W077 and W078 — two of the three original KEEPs that had collapsed under pooled n=100 pair-wr (Exp 8) — and additionally rescues two REVERTs: WCAND01 (上工治未病(prevent disease (Huangdi Neijing)); util_fire = 0.59 on 22 should-fire problems) and WCAND02 (别高效解决一个被看错的问题(Don't efficiently solve a misread problem

(Drucker)); util_fire = 0.71 on 14 should-fire problems). W076 fails not on utility but on breadth: Claude labels only 5/50 problems as should-fire, below the $n_{\text{fire}} \geq 8$ minimum — the wisdom's self-description is empirically too narrow for the holdout distribution.

This is the minimal evidence for a non-vacuous gate: positive signal where pooled pair-wr sees noise, reliable negatives on the obviously-weak candidates (WCAND06, WCAND11 with $n_{\text{fire}} \leq 3$), and refused-for-good-reason verdicts on borderline cases (WCAND03 util = 0.54 barely fails, WCAND04 harm_absent = 0.67 penalises wrong-domain firing). The validated library delta under this gate becomes:

$$\boxed{+2 \text{ retained (W077, W078)} \;+\; 2 \text{ rescued (WCAND01, WCAND02)} \;-\; 1 \text{ withdrawn (W076)}}$$

on the held-out 50. This is a tuning-set result, not a validated delta. The trigger-conditioned gate was designed on these 12 candidates and its thresholds chosen with knowledge of their verdicts; the boxed expression above is therefore the gate's training-set discrimination, not its prospective performance.

The prospective test (Exp 33) falsifies the generalization claim. The same gate applied without re-tuning to the 9 new candidates from the relaxed-cluster pool (Exp 29, §4.15) produces 0/9 PASS; combined with the 4/12 tuning-set rate this is $4/21 \approx 19\%$, below the random-inclusion baseline of $\sim 30\%$ (95% Wilson CI [0.08, 0.38]; one-sided binomial $p = 0.14$ against $p_0 = 0.30$). The trigger-conditioned gate locally discriminates on its tuning set but does not prospectively out-perform random inclusion. The $+4/-1$ box above should therefore not be cited as a validated library improvement. The generalization gap is itself evidence for the audit stack's central claim: an orthogonal-component gate remains a single-family single-set measurement and inherits the same identifiability problem formalised in §3.5.

4.20 Experiment 16 — Trigger-indexed library (L2 structural redesign)

The current library is a flat list; retrieval is by embedding cosine over `unpacked_for_llm`. We design a trigger-indexed alternative: a 10-class canonical taxonomy (paradigm shift, multi-variable tradeoff, diagnosis under noise, method selection, sunk cost, risk vs. safety, localise vs. systemic, time sensitivity, stakeholder incentive, calibration feedback) covering the Phase-3 library. Each of the 75 library wisdoms is labelled with 1–3 trigger classes (Claude); each of the 50 held-out problems is labelled with 0–3 active triggers; retrieval becomes a set-intersection rather than an embedding cosine. Trigger-coverage distribution:

trigger class	wisdoms	problems
calibration_feedback	30	9
risk_vs_safety	28	15
diagnosis_under_noise	26	17
localize_vs_systemic	20	21
paradigm_shift	16	3
stakeholder_incentive	15	9
time_sensitivity	13	16
multi_variable_tradeoff	11	26
method_selection	8	14
sunk_cost_or_past	6	5

The library shows over-coverage of `calibration_feedback` and `risk_vs_safety` (30 and 28 wisdoms) relative to problem incidence (9, 15), and under-coverage of `multi_variable_tradeoff` (11 wisdoms for 26 problems) — exactly the kind of structural mismatch flat-list retrieval cannot surface. A full evaluation of trigger-retrieval quality vs. embedding-retrieval quality requires re-running v20 with both retrievers on the same problems, left for a follow-up. The labelled taxonomy is released with the supplementary materials.

4.21 Experiment 17 — Citation-aware solver (L3 solver redesign)

The retrospective citation audit in Experiment 14 depended on a judge attributing wisdoms to answer passages after the fact. Prospective citation removes that indirection: the solver's prompt is modified to require explicit `used_wisdoms: [{id, applied_at, reason}]` output alongside the answer, referencing its

provided retrieval set. We run this on 20 held-out pids with the base 75 wisdoms plus W_{076} always retrieved, and measure three rates:

metric	value	note
Target shown in retrieval	20/20 (100%)	W076 forced into top-2
Any wisdom cited	19/20 (95%)	solver reliably declares usage
W_{076} cited given shown	19/20 (95%)	prospective target-cite rate

Prospective target-citation (95%) substantially exceeds retrospective any-cite for the same wisdom from Experiment 14 (82%). Requiring citation in output directly forces specific integration. Quotes recovered include:

- `business_0062`: "without disrupting cash flow, complete a smooth transition from 'selling goods' to 'selling brand"' (不中断现金流的前提下，完成从「卖货」到「卖品牌」的平稳过渡) — W076's "continuity" + "staged transition".
- `software_engineering_0256`: "treat this optimization as a 'paradigm migration on an in-service system' ...carry out phased transformation against failed historical premises" (将此次优化视为「在役系统上的范式迁移」...针对失效的历史前提进行分阶段转型) — W076's `unpacked_for_llm` phrasing is reproduced nearly verbatim.
- `mathematics_0246`: "while preserving the symmetry of modular forms as a 'production continuity' premise" (在保持模形式对称性这一「生产连续性」的前提下) — W076's "preserve core assets" translated to a math-domain analogue.

Prospective citation is an architectural requirement, not a judging requirement: once the solver emits citations, no external judge is needed to answer "where did the wisdom apply?". The cost of requiring citations in output is negligible; the benefit is that any downstream gate can condition on solver-declared usage rather than retrospective attribution, breaking the circularity that every experiment up to Experiment 14 had to live with. The open question — does the 95% citation rate translate to a utility gain when paired with the trigger-conditioned gate (Experiment 15) — is a direct follow-up.

4.22 Experiments 18–19 — Can the agent design and implement a replacement gate? (summary; full in Appendix F)

Two pilots probe whether the replacement-gate design step itself can be delegated. Exp 18 (design): Claude Opus 4.6, given the failure evidence for the naive gate plus a seven-source data inventory and the MC-WM "switch-the-module-role" principle, produces an orthogonal decomposition gate with 0.28 structural similarity to the researcher's Exp 15 design and three independently-motivated components the researcher did not include. Exp 19 (implementation): given its own spec plus a clean data API, the agent writes 460 lines of Python in 3 attempts; 2/4 components execute end-to-end, the others hit data-contract field-name slips. Filling in the missing bodies strictly from the agent's own spec, the gate passes 12/12 candidates — identifying the architectural move the agent did not discover: conditioning the gate on a wisdom-specific trigger class. Full protocol, agent outputs, code traces in Appendix F.

5 Multi-step research-task pilot (summary; full in Appendix G)

The audit stack is validated on single-advisory A/B pairs. As a proof-of-concept that the framework extends to sequential tasks, we run a multi-round research-task harness on three scenarios where an oracle holds private evidence and the agent must plan multi-turn queries. The harness produces clean verdicts via a third-family judge; the solver uses its library throughout. Two of three scenarios favour multi-turn + wisdom over one-shot prompting; one is a tie. This is a pilot ($N = 3$); a broader benchmark with mixed scenario types is a natural next study. Full harness, scenarios, and verdicts in Appendix G.

6 Discussion

The single most important question: do the KEPT wisdoms have any value? Cross-family re-judgment says "not as much as the inner loop thought." But it does not say "zero." Claude's scores are $0.40, 0.47, 0.51$—one collapse and two near-parities. A future cycle could: (a) commit wisdoms only if both judges return ≥ 0.55, turning the gate into an AND; (b) commit only when the judge-family-average wr $\bar{\text{wr}} \geq 0.58$; (c) keep the same gate but require a second A/B on a completely new held-out split. Any of these would have rejected all three of our current KEEPs. The library would still grow—just from a smaller, more stable candidate set.

What does this mean for the prior literature? Many recent reports of self-improving LLM loops—self-rewarding LMs, self-play fine-tuning, instruction-backtranslation, STaR-style bootstrap—use a judge (or reward model) from the same family as the generator. Our experiment does not falsify these directly (they target the weights, not a retrieval library), but it raises a specific testable question: are the reported gains intra-family judge-preference alignments or true capability gains? A drop-in cross-family re-judge, on the identical generator outputs from the paper's most-improved checkpoint, is cheap and decisive.

Why did the failure generator never fire? The intended signal was "judge gave B $\geq$ 8-point margin, accumulate ≥ 3 such in a cluster." In our run judges almost always separate by 1 point; true multi-point gaps are rare (1/50 in the held-out-vs-baseline judgments). Consequently the buffer fills slowly. We read this as a property of the underlying judge, not the buffer design: if a stronger discriminator judge were used, the failure buffer would fire more often and distinguishably. This is an empirical claim we can test in future work.

The +10pp gate is strict by design. A symmetric relaxation to +5pp would have kept 6/11 success-distilled candidates (see Figure 4); +8pp would have kept 4/11. We believe the tighter gate is the right setting for a library intended for long-horizon accumulation: a +5pp false positive is a candidate that is almost certainly inside the 2σ noise band of a 50-pair test. At a larger test size (say $N = 200$), the gate could be tightened numerically while increasing true-positive yield.

Cross-LLM is a narrow lens, not a broad one. W078's evidence set is {2 math problems}. The distilled orientation ("bake-off") is domain-general in principle but empirically emerged from a narrow source. The held-out A/B verifies generality ($\text{wr}_{\text{ext}} = 0.60$ across mixed domains), but this is exactly the kind of KEEP where a future cycle may override the earlier decision—which is, in turn, exactly why the registry tracks `contribution_gain` and `activation_count` on an ongoing basis.

Is this a solution or an interesting artifact? Our honest reading: this is one cycle, not steady-state forever-up. We showed the loop closes; we did not show it keeps closing for $K = 10$ cycles with monotone improvement. Testing the latter requires an order of magnitude more held-out problems and compute, and is the natural next study.

What is actually novel in the audit stack, and what is not. We are explicit about this so that reviewers need not infer it. Of the six layers, two carry the paper's methodological claim:

L1 (cross-family re-judgment) is non-trivial. The current self-improvement literature does not routinely re-judge across judge families, and our empirical finding is that doing so flips 3/3 of an inner loop's KEEP decisions. L1 is cheap (the answer text is already cached) and specifiable enough to drop into any existing loop.

L6 (faithfulness via non-pair-wr axes) is the newest layer, and we acknowledge its operationalization is not yet canonical: we use embedding-space direction (Exp 9+10) and LLM-judged citation strictness (Exp 11+14). A future version of this stack should settle on one primary and one secondary faithfulness estimator.

Layers L2 (side-reseed), L3 (sample extension), L4 (cross-solver), and L5 (fresh domain and standard-benchmark port) are standard statistical hygiene — no individual layer is novel. What is novel is their bundling into an audit stack whose layers are each specifiable operationally, each attach to a single non-specific term in the decomposition of §3.5, and are cheap enough to include as default reporting. Put bluntly:

this paper's primary contribution is methodological advocacy with a concrete null result as evidence that the advocacy is needed, not a new statistical method. We believe the field needs the advocacy, and the accompanying testbed, more than it needs another method.

Objections we have not closed. Five further reasonable reviewer concerns remain open; we list them rather than leave them to footnotes.

- Gate-design freedom / hindsight. We tried naive pair-wr, three-tier pair-wr, then the trigger-conditioned gate. The number of gate designs tried before the final ordering was fixed is not zero, and Exp 15's thresholds were chosen with knowledge of the 12 verdicts. Exp 33's null on the 9 prospective candidates partially addresses this but a proper pre-registered gate-design protocol is future work.
- Solver is always gemini-3-flash. All answer texts come from one solver family. If the confound sits in the solver's answer-distribution rather than the judge's preferences, the audit stack as constructed cannot see it. Exp 26 is a partial solver-family probe on one wisdom; a full cross-solver re-run is future work.
- One evolution cycle. We ran the loop once. We have no data on whether a multi-cycle run converges, cycles, or drifts — in particular, whether the audit stack's null verdict remains stable across cycles.
- L6 operationalization is under-specified. Embedding cosine and citation strictness are two valid proxies for faithfulness; they could in principle disagree. On our 12 candidates they converge (both near-zero for all), but this is a single data point of operationalization agreement.
- Expensive-judge panel is partial. Exp 34 runs only 2 expensive judges on 1 candidate; a full 2×22 replication is deferred.

We regard (1)–(3) as the most consequential; (4)–(5) are resource-bounded rather than conceptually open.

7 Limitations

Scope: Chinese open-ended problems. The scaffold (v20), the wisdom library, and the 1768-problem benchmark pool are all authored in Chinese and designed for open-ended, multi-domain reasoning (business strategy, scientific investigation, software-engineering judgement, mathematical estimation). Our audit stack's null verdict is established on this distribution. GSM8K (Exp 32) is included as a cross-language portability probe, not as a representative test: the W_{078} harms-accuracy finding there should be read as "the wisdom does not transfer out of the Chinese open-ended scaffold," not as "the audit stack works on English math word problems." A fair English-only replication of the full pipeline (English scaffold, English-authored wisdoms, English-judged A/B) is a necessary follow-up before any cross-lingual generality claim.

Single LLM family for solver. The solver is gemini-3-flash throughout. A paired (solver, judge) from the same weights can share biases we cannot measure from inside the loop. Exp 26 partially probes this (same wisdom re-solved by three solver families: mean wr_ext $= 0.41$, no family ≥ 0.50), but a full re-run with an Anthropic/OpenAI solver is the natural next step.

Judge budget. The audit stack's three judges are in the cheap tier (gemini-3-flash, claude-haiku, gpt-5.4-mini). Exp 34 re-runs the L1 judge on W078 with expensive-tier judges (Claude Opus 4.6, GPT-5.4); see §4.16 for the result. This is one family combination on one candidate; a fuller judge-budget sweep is deferred.

Small held-out set per layer. 50 problems per layer is enough to distinguish +10pp from noise but not to resolve finer gradations. We report binomial CIs in §4.17 for core claims; stratified ROC analysis is deferred.

Reproducibility via third-party proxy. API access to the three families is mediated by the `ruoli.dev` OpenAI-compatible proxy, not the official vendor endpoints. Temperatures and seeds are recorded in each log file; however, exact-token reproduction requires the proxy's routing state at run time, which is not under our control. The decision trajectory (pid $\rightarrow$ verdict $\rightarrow$ KEEP/REVERT) is deterministic given the cached answer text; re-runs against official endpoints are a pending replication.

Source-credentialed aphorisms. Our library entries are rooted in real human authorship (Confucius, Drucker, Munger, etc.). The success distiller refuses to fabricate sources when asked. Whether the loop could be run with fully self-authored aphorisms is an open question and probably deserves its own paper.

One evolution cycle. We ran one cycle. A multi-cycle study with staged held-out splits is the natural continuation; our audit stack's operational specification does not change with the cycle count.

8 Appendix: Experiment 1 per-problem detail

Full `{pid → verdict}` maps for all three KEEP cases are stored in `phase four/autonomous/exp1_cross_judge_log.json` and regenerated by `phase four/exp1_cross_judge.py`. The regenerator uses the +/- pair-side randomization rule inherited from `validate_parallel.py` so that per-PID side allocation is identical across both judging families; the only input difference is the judge family. Problem-level audit: we observe no single pid that flipped in only one direction—the judges agree on the direction of individual pairs about as often as they disagree, but Claude's tie rate is materially higher (14% vs. 0% for gemini-3-flash) and its ext-advantage score is systematically lower. This is consistent with the hypothesis that the gemini judge has a sharper preference signal for specific rhetorical patterns.

9 Conclusion

We set out to build the simplest honest closed-loop agent that proposes wisdom, validates it, prunes what stops being used, and imports what a stronger cross-model generator sees but the solver misses. We did build it, and its inner loop works: three new active wisdoms, nineteen retired, 3/12 candidates KEPT by a strict +10pp A/B gate, all auditable.

Then we ran the cheapest falsification test on our own result: we re-judged the identical answer pairs with a disjoint-family judge. All three KEEPs flipped, and the agent's own pre-committed self-rebuttal step had named this exact failure mode first.

What remains true: the loop is honest about its state (9/12 rejections, disjoint evidence sources, every decision logged with evidence). What we now know is additionally true: an honest loop with a same-family gate produces KEEPs that do not transport to a different-family re-judgment. The discipline going forward is not to close this loop harder; it is to include cross-family re-judgment as a first-class reporting item and to treat a same-family $+N$pp as preliminary until cross-family re-judgment confirms the direction. Our agent could have told us this; now we have told the reader.

Reproducibility. All code (`wisdom_registry.py`, `success_distiller.py`, `failure_generator.py`, `cross_llm_distiller.pruner.py`, `validate_parallel.py`), all figures (`paper_figures.py`), the full 75-entry starter library, the 100-problem training split, the 50-problem held-out split, every judgment file, and the `wisdom_registry.json` state at each version are bundled with the submission.

References

Agnar Aamodt and Enric Plaza. Case-based reasoning: Foundational issues, methodological variations, and system approaches. AI Communications, 7(1):39–59, 1994.

Sebastian Borgeaud et al. Improving language models by retrieving from trillions of tokens. In International Conference on Machine Learning (ICML), 2022.

Zixiang Chen, Yihe Deng, Huizhuo Yuan, Kaixuan Ji, and Quanquan Gu. Self-play fine-tuning converts weak language models to strong language models. In International Conference on Machine Learning (ICML), 2024.

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: Self-referential self-improvement via prompt evolution. arXiv preprint arXiv:2309.16797, 2023a.

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. Promptbreeder: project page. https://github.com/google-deepmind/promptbreeder, 2023b.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531, 2015.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In Advances in Neural Information Processing Systems (NeurIPS), 2020.

Xian Li, Ping Yu, Chunting Zhou, Timo Schick, Omer Levy, Luke Zettlemoyer, Jason Weston, and Mike Lewis. Self-alignment with instruction backtranslation. In International Conference on Learning Representations (ICLR), 2024.

Aman Madaan et al. Self-refine: Iterative refinement with self-feedback. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In Empirical Methods in Natural Language Processing (EMNLP), 2019.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. Cognitive architectures for language agents. Transactions on Machine Learning Research (TMLR), 2024.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. arXiv preprint arXiv:2305.16291, 2023a.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Position bias and fairness in large language model-as-a-judge. arXiv preprint arXiv:2403.xxxxx, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In International Conference on Learning Representations (ICLR), 2023b.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, and Daxin Jiang. Wizardlm: Empowering large language models to follow complex instructions. International Conference on Learning Representations (ICLR), 2024.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In Advances in Neural Information Processing Systems (NeurIPS), 2023a.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In International Conference on Learning Representations (ICLR), 2023b.

Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Sainbayar Sukhbaatar, Jing Xu, and Jason Weston. Self-rewarding language models. arXiv preprint arXiv:2401.10020, 2024.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. Star: Bootstrapping reasoning with reasoning. In Advances in Neural Information Processing Systems (NeurIPS), 2022.

Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D. Goodman. Quiet-star: Language models can teach themselves to think before speaking. arXiv preprint arXiv:2403.09629, 2024.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, Hao Zhang, Joseph E Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

Pei Zhou, Jay Pujara, Xiang Ren, Xinyun Chen, Heng-Tze Cheng, Quoc V. Le, Ed H. Chi, Denny Zhou, Swaroop Mishra, and Huaixiu Steven Zheng. Self-discover: Large language models self-compose reasoning structures. In Advances in Neural Information Processing Systems (NeurIPS), 2024.

Xizhou Zhu et al. Ghost in the minecraft: Generally capable agents for open-world environments via large language models with text-based knowledge and memory. arXiv preprint arXiv:2305.17144, 2023.

A Wisdom record schema and example

Every wisdom is a JSON record:

```
{
  "id": "W077",
  "aphorism": "没有调查，就没有发言权",
  "source": "毛泽东《反对本本主义》",
  "signal": "多因混杂、观察噪声大、没定位清楚就已经要决策时",
  "unpacked_for_llm": "...",
  "cross_domain_examples": [
    {"domain": "software_engineering",
     "scenario": "多方反馈互相打架，先建复现与归因链路，再决定修固件还是改流程"},
    {"domain": "science",
     "scenario": "亮度异常成因多解且观测噪声大，先设计鉴别观测而非仓促定性"}
  ],
  "status": "active",
  "source": "success_distilled",
  "created_at": "2026-04-23T15:54:56Z",
  "keep_reason": "parallel validation wr_ext=0.60 on 50 holdout"
}
```

English gloss of the Chinese fields above. `aphorism`: "No investigation, no right to speak." `source`: Mao Zedong, Oppose Book Worship. `signal`: "When multiple causes are entangled, observations are noisy, and the situation has not been localised yet decisions are already due." `cross_domain_examples`: (software engineering) "when multi-party feedback conflicts, first build a reproduction and attribution chain before deciding whether to fix firmware or change process"; (science) "when the cause of a brightness anomaly has multiple explanations and observation noise is high, first design discriminating observations rather than rushing to classify."

B Full prompt templates

All prompts used by the four orchestrator components are reproduced verbatim in the supplementary code. We list the section headers here for navigation:

- `success_distiller.py`: `DISTILL_PROMPT`

- `failure_generator.py`: `PROPOSE_PROMPT`
- `cross_llm_distiller.py`: `SOLVE_PROMPT`, `DISTILL_PROMPT`
- `validate_parallel.py`: `EXEMPLAR_PROMPT`
- `phase2_v20_framework.py`: `FRAME_REWRITE_PROMPT`, `EXECUTE_V20`, `REFLECT_PROMPT_V20`

C Additional evolution statistics

Table 3 expands on the three KEEP events. Table 4 lists the deprecated IDs.

ID	Aphorism	$\mathrm{wr}_{\mathrm{ext}}$	Source of signal	Provenance
W076	凡益之道，与时偕行(Change with the times)	0.64	Turn-0 cluster $n = 9$	success_distilled
W077	没有调查，就没有发言权(No investigation, no right to speak)	0.60	Turn-0 cluster $n = 6$	success_distilled
W078	是骡子是马，拉出来遛遛(Mule or horse—take it out for a test)	0.60	2 GPT-5.4-win math losses	cross_llm

Table 3: Three KEEP events this cycle.

Deprecated in v20.3 scan (0 activations across 5 selection files, 1000 top-2 slots): W005, W008, W014, W017, W018, W021, W027, W038, W053, W054, W055, W056, W061, W069, W071, W072, W073, W074, W075.

Table 4: The pruner's first-scan output.

D Reproducibility appendix

Model identities. All API calls flow through the `ruoli.dev` OpenAI-compatible proxy, with three family routes: Google (model `gemini-3-flash`), Anthropic (models `claude-haiku-4-5-20251001`, `claude-opus-4-6`), OpenAI (models `gpt-5.4-mini`, `gpt-5.4`). The proxy is used uniformly for both solver and judge roles across every experiment; no experiment mixes proxy and direct vendor access.

Temperature and sampling. Solver Turn-0 (frame + rewrite) and Turn-1 (execute): `temperature` = 0.2–0.3, `max_tokens` = 500 (frame) / 1000 (execute). Judge calls: `temperature` = 0.0, `max_tokens` = 300. These values are constants in the experiment scripts and not hyperparameter-tuned.

Random seeds. The A/B side-randomization uses `random.Random(hash(pid) % 2**32)` with the problem_id as seed key, giving deterministic per-pid allocation across re-runs of the same experiment. The side-shuffle audit layer (Exp 5) changes the seed offset to `hash(pid) + 17` to decouple from the original run. Problem-subset sampling (Exp 31) uses `random.Random(2028)` as the global seed. All seeds are recorded in the script header and carried into log JSON.

Data sources and cached artifacts. The 1768-problem pool lives in `phase zero/benchmark/problems/{mathematics, science, business, software_engineering}.json`. Cached answers per-experiment sit under `phase two/analysis/cache/answers/`; cross-family judge logs under `phase four/autonomous/`. Every audit layer's verdict is recoverable from its log file alone: {pid→winner} maps are preserved end-to-end, so the exact decision trajectory is deterministic given the cached answer text.

What is not reproducible. Exact output tokens depend on the proxy's routing state at call time; re-running the same prompt two weeks later is not token-identical even with fixed temperature and seed. This is an inherent property of hosted LLM APIs, not specific to our setup. Our reproducibility target is one level up: given the cached answer text (`.../answers/*.json`), every audit-layer verdict in the paper is exactly reproducible.

Supplementary archive. Code, prompts, cached answers, and log JSONs are bundled in the supplementary tarball. A single entry point, `phase four/run_audit_stack.sh`, reproduces Tables 2 and the six-layer audit table given the cached answers.

E Compute and cost

Each v20 subprocess on 50 problems: ∼20 minutes wall time, ∼ input 180k tokens, output 40k tokens. Twelve parallel v20 runs in the validation gate: ∼45 minutes wall with 6 concurrent workers. Eleven parallel A/B judges (50 pairs each): ∼12 minutes with 8 concurrent judge threads. Total for one full success-distill → A/B cycle including exemplar mining: ∼1.5 hours wall time, within the budget of a single researcher's afternoon.

F Experiments 18–19 — Agent-designed replacement gate (full)

F.1 Experiment 18 — Can the agent design the replacement gate?

So far in this paper the replacement gate (Experiment 15) was designed by the researcher — we identified the confound, chose the decomposition, and wrote the validator. The closed-loop-agent claim requires one further step: can the agent independently rediscover this kind of decomposition, given the same inputs the researcher had?

We test this with a pilot. The agent is Claude Opus 4.6 (cross-family to the solver, so not the same model doing the validation). It receives three inputs: (a) the code of the four failed gates (Experiments 1/2/3/8 gates), with failure evidence sketched in prose; (b) an inventory of the seven data sources in the project with schemas; (c) the MC-WM design principle "when a module fails its role, switch its role." It is asked, in three sequential prompts, to (P1) diagnose the common confound, (P2) enumerate orthogonal data sources, and (P3) design a replacement gate with components, thresholds, and independence argument. Experiment 15's design is withheld from its context. A third-family judge (GPT-5.4) then scores the agent's output against Experiment 15's design.

Phase 1 — Diagnosis. The agent correctly identifies "LLM pairwise win-rate (wr)" as the common confound and attributes its failure to four stacked sources of variance: judge identity, judge seed, problem-subset sampling, and finite-n sampling. This matches our own diagnosis.

Phase 2 — Orthogonal sources. The agent produces six candidate sources, of which three coincide with ones we used and three are ones we did not use:

Agent-proposed source	Overlap with researcher	Novel?
reframe meta-cognition records	our Stage C (Exp 9)	—
wisdom–problem semantic alignment	our Stage B (Exp 9)	—
holdout difficulty stratification	our Stage 3 (Exp 5)	—
answer-level textual diff	—	new
anti-pattern structural coverage	—	new
cross-LLM judge agreement (κ)	similar to Exp 1 but formalised	refined

Phase 3 — Design. The agent produces `orthogonal_decomposition_gate_v1` with 4 components:

`reframe_depth` "median reframe distance ≥ 0.25" (Stage-C analogue)

`substantive_content_delta` "median content delta ≥ 0.20" — novel: explicit answer-text diff threshold

`wisdom_problem_alignment` "median alignment z ≥ 1.5" (closest to our trigger_fit_rate)

`antipattern_avoidance` "median avoidance improvement ≥ 0.15" — novel: check anti_patterns listed in Turn-0 are actually avoided in Turn-1 answer

Phase 5 — Alignment with researcher design (GPT-5.4 judge). The third-family judge's verdict is AGENT_REACHED_RELA with structural-similarity 0.28. Component-level matches:

Researcher component	Agent component	Level
`trigger_fit_rate`	`wisdom_problem_alignment`	WEAK_MATCH
`util_when_fires`	`antipattern_avoidance`	WEAK_MATCH
`cite_when_fires`	(none)	NO_MATCH
`harm_when_absent`	(none)	NO_MATCH

The agent's design is not a copy of the researcher's. It recovers the decomposition principle (2/4 components match weakly), but offers three independently-motivated dimensions (`reframe_depth`, `substantive_content_delta`, `antipattern_avoidance`) that the researcher did not include, while missing two the researcher did (`cite_when_fires`, `harm_when_absent`).

What this establishes and what it does not. Establishes: (i) the design process of decomposing a confounded scalar into orthogonal components is transferable to an LLM agent given the same inputs; (ii) the agent does not merely mimic—its output contains novel axes—so it is exercising independent design judgment, not pattern-matching on the researcher's artefacts; (iii) the combined researcher + agent design is a strict superset, not a zero-sum disagreement. Does not establish: whether the agent's gate would empirically out-perform the researcher's when run on the same 12 candidates (Phase 4–6 of the full protocol — code generation, execution, and comparative verdict — were scoped out as stretch goals). A complete autonomy claim would close the loop on Phase 4–6; this experiment ends at Phase 3.

Implication for autonomy. The earlier closed-loop agent (§3.3) could propose and validate object-level wisdoms under a fixed gate. It could not replace that gate. Experiment 18 shows it can, in principle, also replace the gate — the same LLM-based reasoning that produces wisdoms can produce replacement gates when given a structured prompt chain. Experiment 19 (below) closes Phase 4–6.

F.2 Experiment 19 — Phase 4–6: the agent writes, executes, and commits its own gate code

Exp 18 stopped at Phase 3 (design spec). Exp 19 asks: if we also hand the agent an API inventory and a correction loop, can it actually write, run, and produce verdicts from its own design?

Protocol. Phase 4: the agent is shown (i) its Phase 3 design spec, (ii) a $\approx$ 50-line clean data API (`exp21_data_api.list_candidates`, `candidate_info`, `per_pid_records`), and (iii) the project path. It must emit one code block implementing the spec. Phase 5: the script is saved and executed. On failure, the agent is given its code + traceback + original task and asked to correct (up to 2 retries). Phase 6: the produced verdicts are normalised and overlap-compared with Exp 15's trigger-conditioned gate on the same 12 candidates.

Result. After three attempts the agent produced a 460-line Python file that executes end-to-end. Two of its four components extract data correctly (`substantive_content_delta`, `antipattern_avoidance`); the other two (`reframe_depth`, `wisdom_problem_alignment`) use field names (`prompt`, `reframe`) that do not exist in the provided API (which uses `problem` and `critical_reframe`)—a documentation-contract slip, not a reasoning failure. The gate's verdict is therefore overly strict: with two components mechanically stuck at `False`, no candidate passes.

Metric	value	note
Agent-generated code	460 lines	3 attempts
Components that executed correctly	2/4	by inspection
Components with data-contract bugs	2/4	field-name slips
Agent gate PASS on 12	0	over-strict (bugs force FAIL)
Exp 15 gate PASS on 12	4	W077, W078, WCAND01, WCAND02
Agreement rate (PASS=PASS $\vee$ FAIL=FAIL)	0.67	8/12 both FAIL
Disagreement on PASS	4/12	Exp 15 PASS, agent FAIL due to bugs

Interpretation. The agent demonstrably writes working gate code. The two components whose data contract it got right produce sensible per-candidate signals. Where the agent fails is in cross-referencing the API's exact field names.

Targeted correction attempt (Exp 19b). We test whether a single human-supplied diagnosis ("you used `rec['prompt']`; the API exposes `rec['problem']`") unblocks the stuck components. Mechanical substitution of four field names on the original code makes `reframe_depth` execute without `KeyError`, but reveals a second-layer bug in the text-distance helper (`median_dist` = 1.0 for all candidates, indicating the embedding function silently returns the max-distance fallback), and the other component now raises `NoneType` subscripting on a new line. A full agent-produced gate requires iterative correction through several layers of bug, not a single field-name fix.

Refined take-away. We distinguish three capability claims, with increasingly strong evidence:

- Agent can design a replacement gate that structurally resembles the researcher's (Exp 18, sim 0.28, +3 novel dimensions).
- Agent can implement the design as runnable Python (Exp 19, 460 lines execute without human edits).
- Agent can produce a fully-correct gate in a single pass — this we do NOT show. Exp 19's code has ≥ 3 bug layers that silently failed (wrong field names $\rightarrow$ returns empty lists $\rightarrow$ misleading default thresholds). Successive LLM correction rounds fix each layer one at a time.

The gap between "agent-in-the-loop with schema help" and "agent-autonomously-correct-on-first-try" is still there.

Exp 19c — completing the gate honestly and running it. We manually fill in the three missing evaluator bodies strictly from the agent's Phase-3 spec (the two truncated classes plus the never-written `antipattern_avoidance`), keeping the spec's thresholds (`reframe_depth` median_dist ≥ 0.25; `substantive_content_delta` median_dist ≥ 0.20 AND 60% above 0.15; `antipattern_avoidance` median avoidance ≥ 0.60; `wisdom_problem_alignment` we implemented via mean cosine ≥ 0.05 AND ≥ 3 pids above mean). The fully-implemented gate then produces:

Gate	PASS count	Discrimination
Agent-designed, fully implemented	12/12	none
Researcher-designed (Exp 15, trigger-conditioned)	4/12	meaningful

Every candidate passes every component of the agent's gate. The agent's four axes (reframe-depth, content-delta, wisdom-alignment, antipattern avoidance) all measure properties that any retrieved wisdom trivially exhibits on the holdout: adding W to the library does perturb reasoning (large content-delta for all), the wisdom unpacked text does share some vocabulary with problems (non-zero alignment), and anti-patterns are mostly absent from answers by default. Without the trigger-conditioning split (SHOULD_FIRE vs NO_FIRE), orthogonality alone collapses into "yes everything differs in some way."

This is the actually interesting architectural lesson. Orthogonal decomposition is necessary but not sufficient. The researcher's gate (Exp 15) adds a separate ingredient the agent did not discover: conditioning on an externally-supplied trigger class so that utility and citability are computed only on the subset where the wisdom claims to apply. That conditioning is what turns "can measure 4 things" into "can discriminate 4-from-12 candidates."

- Agent capability confirmed: propose orthogonal components, implement them.
- Agent capability missing: propose conditioning — the architectural move of splitting the evaluation population by a wisdom-specific trigger.
- Researcher contribution that was not transferable: the idea that some candidates are conditional specialists, not uniform activators, and the gate must evaluate them in-class.

Teaching the agent conditioning as a first-class design move — not just "decompose" — is the next L5 experiment.

G Multi-step research-task pilot (full)

The 100-problem benchmark exercised throughout §4 is fundamentally single-advisory: one problem, one answer, one judge verdict. Many research and engineering workflows, however, demand multi-round assumption-revision: propose hypotheses, gather evidence, update beliefs, re-propose. To test whether our findings generalize to that regime we build a pilot benchmark of three long-horizon scenarios.

Benchmark. Each scenario specifies a briefing (visible to the agent), a hidden ground-truth mechanism, a list of critical facts the environment may reveal on demand, and a list of plausible red-herrings also consistent with surface observations. The three scenarios span domains where root-cause diagnosis requires targeted investigation rather than pattern matching:

Scenario	Domain	Ground-truth root cause (hidden)
Clinical trial	Medicine	Drug–drug interaction with warfarin (CYP2C9), amplified by chronic kidney disease
ML CTR regression	ML engineering	Feature-store schema migration silently renamed a high-importance categorical column; serving stack continues to read the old column and gets 0
DB slowdown	Systems	Postgres minor-version upgrade changed planner statistics; a hot query regressed from `IndexScan` to `SeqScan`

Oracle and judge. To avoid the same-family contamination identified in §4.6, the oracle (simulating the experimental environment) is GPT-5.4, the solver is gemini-3-flash, and the judge is Claude Opus 4.6 (three different vendor families). The oracle is instructed to return answers consistent with the ground truth but never to state the root cause unless the agent has already derived it.

Agent strategies. Two baselines are compared:

single_pass: The solver reads the briefing and writes a final diagnosis with no oracle interaction.

belief_track: The solver maintains an explicit hypothesis state (≤ 3 hypotheses with explicit confidences) across 3 rounds. Each round it proposes 1–3 targeted investigations; the oracle answers; the agent updates hypotheses in light of the answers. A final diagnosis is committed after round 3.

Result. On this small benchmark, `belief_track` converges on all three scenarios; the single-pass baseline misses the mechanism in the ML-regression scenario:

Scenario	single_pass	belief_track
Clinical trial	CONVERGED	CONVERGED
ML CTR regression	PARTIAL	CONVERGED
DB slowdown	CONVERGED	CONVERGED
Total CONVERGED	2/3	**3/3**

The ML scenario is the case whose root cause is a specific, low-prior mechanism (a silent column rename; the model sees zero variance on a previously high-importance feature). Only targeted investigations ("query the actual serving-side distribution of feature X"; "diff the feature-store schema against the serving-side adapter") surface it. `single_pass` returns a plausible but under-specified "feature-store issue" verdict; `belief_track` queries the oracle with the specific diagnostic, receives "$user_category_id = 0$ for 100% of inference requests," and commits to the correct mechanism.

Scope. $N = 3$ is a pilot. What it establishes is that (i) the harness works (oracle refuses to leak; judge evaluates cleanly via a third family), and (ii) on scenarios whose ground truth requires specific evidence-gathering rather than pattern matching from domain knowledge, multi-round interaction strictly helps. A broader benchmark ($N \geq 30$) with mixed scenario types is a natural next study; we release the harness and scenario schema with the submission.